An

Assignment 2 of 2 Report

On

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Submitted By

Youssef Rabie Elkhouly

ID:2510293

British Applied College
الكلية البريطانية التطبيقية

Department of Computer Science

British Applied College, Umm Al Quwain

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Introduction:

Designing/Developing/Building an efficient Network

These Days Modern healthcare facilities depend heavily on reliable and secure network systems to support daily operations and patient care, From managing patient records to enabling communication between doctors, nurses, and administrative staff, a well designed network plays a critical role in ensuring efficiency and safety within a hospital environment.

As a **Network Support Trainee** at **Global Connect Systems LLC**, **I** am involved in designing and deploying a campus wide network infrastructure for a newly established healthcare facility located across three floors, The facility includes multiple departments such as outpatient, inpatient, diagnostics, and administration, all of which rely on networked computer systems, printers, and a central server to operate effectively.

The healthcare facility requires a structured and secure network that can support **80 computer systems**, a **central server** for patient records, and shared network printers across different departments. To meet these requirements, the network design focuses on **automatic IP address management using DHCP** and logical network separation using **VLANs**. These technologies help reduce configuration errors, improve network performance, and enhance data security between clinical and non clinical departments.

For the initial design, the network solution also considers long term **maintenance**, **scalability**, and **security**, Key aspects such as bandwidth management, future expansion, secure access to internal services, and system monitoring are addressed to ensure the network remains reliable as the hospital grows. User feedback from hospital staff is also taken into account to refine the design and improve overall system efficiency, ensuring the network supports both current operations and future needs.

(Burgess, 2004)

Designing an Efficient Network Infrastructure to meet the needs

The healthcare facility oPerates across three floors and supports multiple departments including outpatient services, inpatient care, diagnostics, and administration, Due to the sensitive nature of medical data and the high number of connected systems, the network must be secure, reliable, and easy to manage. The proposed network design follows a campus style layout where all floors are connected through a central networking core, allowing controlled communication between departments.

A hierarchical network structure is used, consisting of a core router and managed switches installed on each floor. This approach improves performance, simplifies troubleshooting, and allows future expansion without major redesign.

How the network Should be ?

Each floor of the building is connected to the core network using structured cabling and managed access switches, and to meet the needs we should follow correct format like:

Ground Floor

The ground floor includes the reception area, administrative offices, and network printers. Systems on this floor are mainly used for patient registration, billing, and office operations. A managed access switch connects all devices and forwards traffic to the core network.

First Floor

The first floor houses doctor's offices, management cabins, and the central server room. This floor plays a important role in the network as it hosts the main server used for **patient records** and hospital management systems. A high capacity switch is installed here to support server traffic and ensure stable connectivity.

Second Floor

The second floor is dedicated to treatment units and nursing stations. These systems require fast and reliable access to patient data. A separate access switch connects all clinical devices to the network, ensuring smooth data flow during medical operations.

VLAN Design?

To keep the network design simple and easy to manage, in the healthcare facility I used a **floor based VLAN** approach, so Each floor is assigned a single VLAN that includes all systems and network devices located on that floor, this method reduces configuration **complexity** while still providing logical separation between different operational areas of the hospital. Since each floor serves a different function, separating them into individual VLANs helps control broadcast traffic and improves overall network performance. *(Donahue, 2011)*

The **ground floor VLAN** includes administrative systems, reception desks, and network printers. The **first floor VLAN** supports doctor offices, management systems, and the central server room, making it a critical part of the network. **The second floor VLAN** is dedicated to treatment units and nursing stations, where reliable and fast access to data is essential. Inter-VLAN communication is controlled by the core router to ensure secure and organised data flow between floors when required.

VLAN Name	VLANID	Floor	Connected Systems
Ground Floor VLAN	10	1	Reception systems, administrative PCs, network printers
First Floor VLAN	20	2	Doctor’s PCs, management systems, central server
Second Floor VLAN	30	3	Treatment units, nursing station systems

DHCP Usage?

Dynamic Host Configuration Protocol also was implemented to automatically assign IP addresses to all connected devices, each VLAN is assigned a unique IP subnet,which helps organise the network and makes fault identification easier, dhcp reduces the need for manual configuration and allows new devices to be added without disruption.

The server VLAN uses static IP addressing to ensure consistent access to hospital services such as patient record systems, internal web services, and file transfers. *(Donahue, 2011)*

Security?

It's a **critical requirement** in a healthcare environment, VLAN separation helps limit access between departments, while the central router controls inter VLAN communication using access rules. This ensures that administrative systems cannot directly access clinical or server resources unless permitted.

When I used managed switches and structured cabling, that **improves performance** and **reduces packet collisions**. This design supports high data traffic from treatment units while maintaining stable access for administrative users.

Topology Used?

I designed the hospital using **Hybrid star topology**, in some cases we can use the name "tree topology", because tree is basically the same family as hierarchical star or the extended star, this topology is suitable for a multi floor healthcare facility because it provides a clear and organised structure while maintaining high reliability, In this design, a central core switch acts as the main connection point for the entire network, and each floor is connected to this core using its own access switch.

Its core switch is located in the first floor server room, where the central hospital server is also placed. This allows fast and stable communication between servers and end user systems, Each floor's access switch connects all local devices such as computers and printers and forwards traffic to the core switch through high speed uplink connections. If a fault occurs on one floor, the issue is isolated to that floor only, which is very important in a hospital environment where continuous operation is required.

Simply this topology supports **VLAN implementation** and inter VLAN routing efficiently, Trunk links between the access switches and the core switch allow multiple VLANs to pass through a single physical connection. As the hospital grows, additional systems or switches can be added without major changes to the existing network design, so the extended star topology provides a balance between performance and scalability and ease of management, making it a suitable choice for the healthcare facility.

(Donahue, 2011)

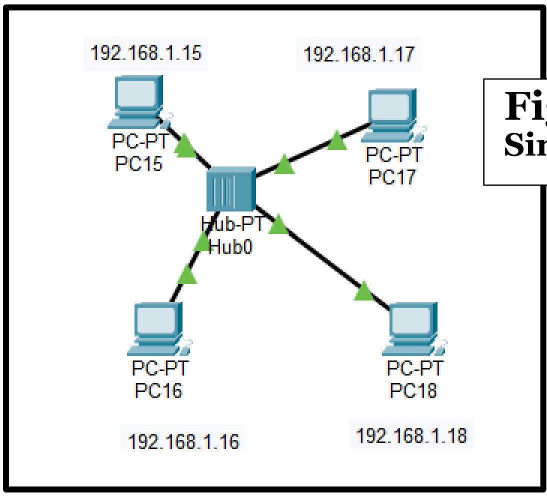


Figure 1:
Simple star topology

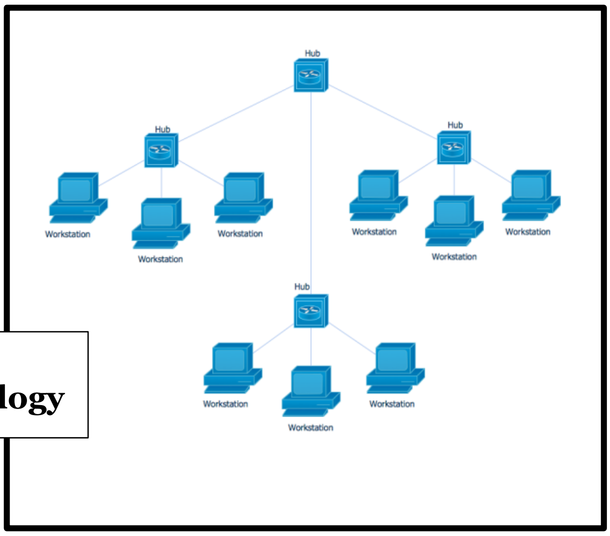


Figure 2:
Hybrid star topology

The Physical design implemented using Packet Tracer?

Figure 3:
The first design

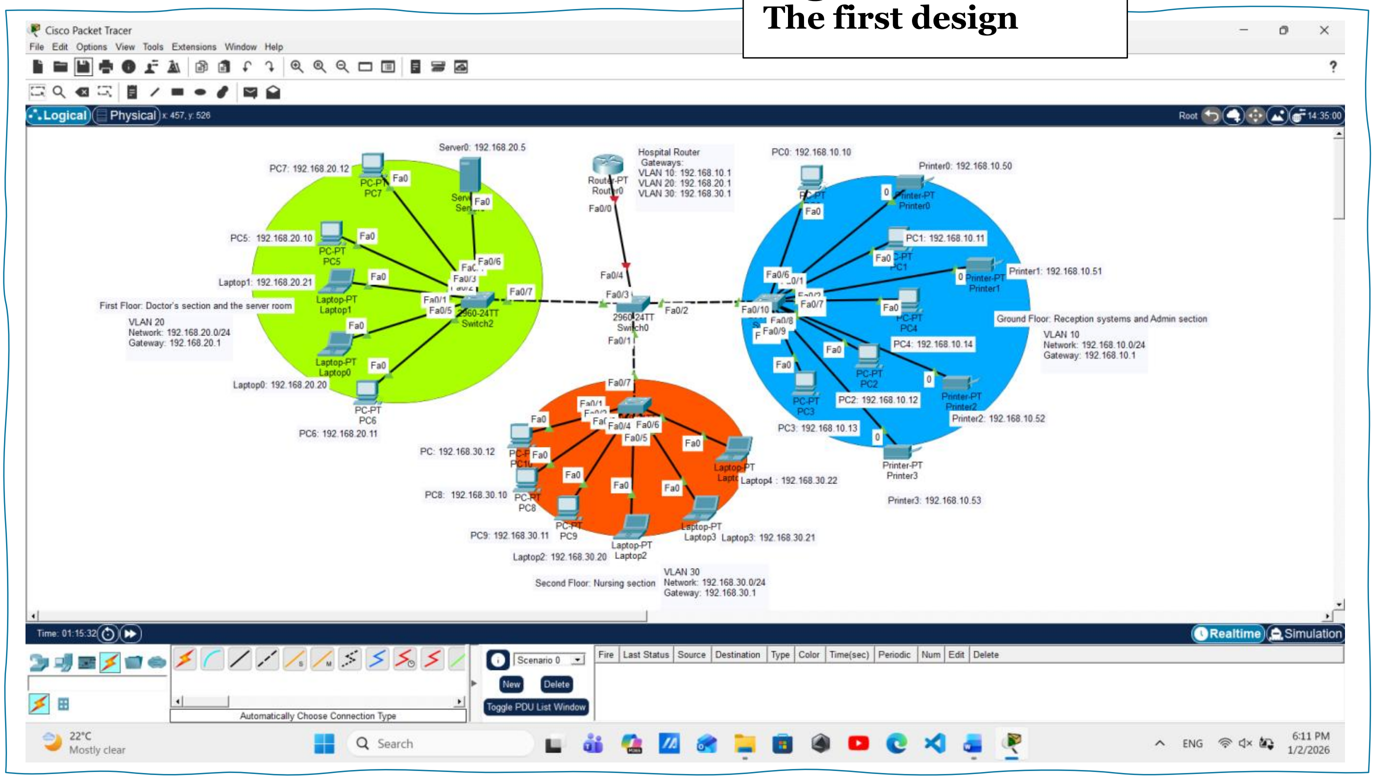


Figure 3 shows the First network design for the healthcare facility, implemented using an hybrid star topology. The network is divided across three floors, with a core switch placed at the center to connect all floor switches, and a hospital router used to provide inter VLAN routing and act as the default gateway, Each floor is assigned its own VLAN and IP subnet to keep traffic organised and secure. The ground floor supports reception and administrative systems with shared printers, the first floor hosts doctor's systems and the central server, and the second floor is used by nursing and treatment units. This design reduces broadcast traffic, improves performance, and allows easy management and future expansion of the hospital network.

Floor	VLAN ID	Staff / Users	Main Devices Used	IP Address Range	Gateway
Ground Floor	**VLAN 10**	Reception staff, Administrative staff	Desktop PCs, Network Printers	192.168.10.2 to 192.168.10.11	192.168.10.1
First Floor	**VLAN 20**	Doctors, Medical consultants, Management staff	Desktop PCs, Laptops, Central Server	192.168.20.2 to 192.168.20.25	192.168.20.1
Second Floor	**VLAN 30**	Nurses, Treatment unit staff	Desktop PCs, Laptops	192.168.30.2 to 192.168.30.25	192.168.30.1

Device	Role in Network	Location	Notes
Hospital Router	Provides inter VLAN routing and acts as default gateway for all VLANs	First Floor (Server Room)	Handles communication between VLAN 10, 20, and 30
Core Switch	Central distribution switch connecting all floor switches	First Floor (Server Room)	Uses trunk links to carry all VLAN traffic

(Göransson and Black, 2014)

How the Network Design Meets the Hospital Requirements

The proposed network design meets the hospital requirements by providing a structured, secure, and scalable solution across all three floors. Each floor is assigned its own VLAN and IP subnet, which helps separate traffic, reduce broadcast congestion, and improve security between departments. The use of an extended star topology with a core switch and a central router ensures reliable connectivity and controlled communication between floors. Automatic IP assignment is planned using DHCP to reduce manual configuration errors, while critical devices such as the hospital server are allocated fixed IP addresses for stable access. Overall, the design supports administrative operations, clinical activities, and patient care systems while allowing future expansion of devices and services.

Hospital Requirement	Design Solution Implemented
Support for 80 computer systems	Scalable extended star topology with floor-based switches
Secure separation between departments	VLAN-based network segmentation per floor
Central storage for patient records	Dedicated server placed in first-floor server room
Automatic IP address management	DHCP planned for all VLANs
Reliable inter-floor communication	Core switch and router providing inter-VLAN routing
Future expansion capability	/24 subnets and modular switch design

(Donahue, 2011)

Keeping the Hospital Network Reliable and Future Ready: Maintenance

Maintenance?

Pointedly this refers to the regular activities carried out to ensure that systems continue to work efficiently and without interruption. In an organisation like a hospital, maintenance is not optional because network failure can directly affect patient care and daily operations. Maintenance helps reduce unexpected breakdowns, improves system reliability, and ensures that both hardware and software continue to perform as expected. By following a planned maintenance approach, issues can be identified early and fixed before they become major problems.

Maintenance in Computer Networks?

In computer networks, maintenance focuses on **performance**, **security**, and **long term usability**. This includes monitoring network traffic, checking device health, updating systems, and planning for future expansion, Since hospital networks handle sensitive patient information and support critical medical systems, maintenance must be done carefully and regularly,because A well maintained network ensures fast data access for doctors and nurses, protects confidential data, and supports the continuous operation of servers and network services. Network maintenance also helps the organisation adapt to growth, such as adding more systems in treatment units or administrative areas. ***(Donahue, 2011)***

Real maintenance schedule “Initial version”

Maintenance Category	Activity Performed	Responsible Area	Time Interval	Expected Outcome
Network Performance	Monitor traffic levels on each VLAN and floor	IT Support Team	**Weekly**	Balanced bandwidth usage and reduced congestion
Hardware Reliability	Check router, core switch, and access switch status	Network Administrator	**Weekly**	Early detection of hardware faults

Data Protection	Verify server backups and storage availability	System Administrator	**Monthly**	Prevention of data loss
Security Management	Review firewall rules, VLAN access, and apply updates	Network Security Team	**Monthly**	Improved protection against threats
Capacity Planning	Analyse network load and device count growth	IT Management	**Quarterly**	Readiness for future expansion
Configuration Review	Audit VLANs, IP addressing, and routing settings	Network Administrator	**Quarterly**	Consistent and organised network configuration

This maintenance schedule is designed to support the hospital's **operational needs** by focusing on performance, security, and scalability, Weekly monitoring ensures that critical areas such as treatment units and nursing stations always have sufficient bandwidth, Monthly security and backup checks protect sensitive patient data and reduce the risk of system failure. Quarterly reviews allow the hospital to plan for future growth without major changes to the network design, so this maintenance approach helps keep the network stable, secure, and capable of supporting hospital operations over the long term.

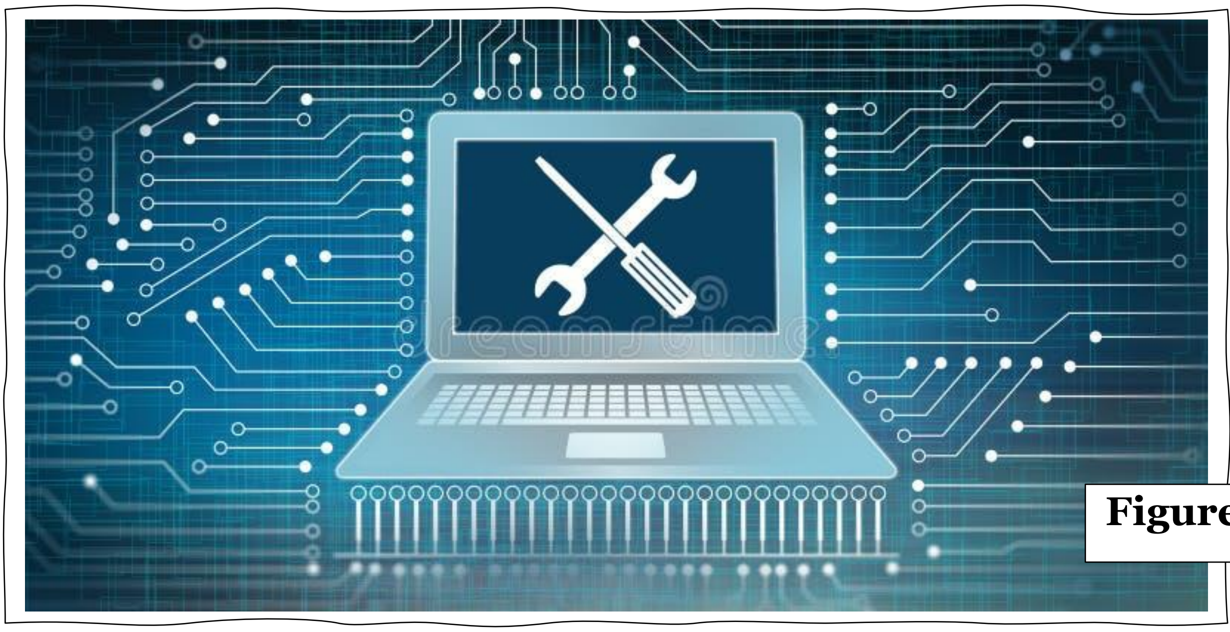


Figure 4

Listening to Users to Improve the Hospital Network: Feedback section

What is feedback?

Modernly it is the information and opinions given by users after they interact with a system. In a hospital environment, feedback from staff such as doctors, nurses, and administrative employees is very important because they use the network daily, Their experience helps identify real problems like slow access, connection issues, or security concerns that may not be visible during technical testing. Feedback allows the network design to be improved based on actual user needs.

What are Surveys?

Surveys are a common method used to collect feedback from multiple users in a structured way, They usually consist of questions that focus on performance, reliability, ease of use, and security. Surveys make it easier to compare responses, identify common issues, and analyse trends across different user groups. In this case, surveys help understand how well the hospital network supports daily tasks across different departments.

How I collected Feedback?

Feedback was collected using an online survey created through MS Teams Forms and shared with hospital staff from different departments. The survey was completed by doctors, nurses, and administrative staff who regularly use the network for their daily work. A total of five responses were collected and used for analysis.

My survey link	https://forms.microsoft.com/Pages/ResponsePage.aspx?id=DQSIkWdsW0yxEjajBLZtrQAAAAAAAAAAAAN__hsIr8pUOVgxR0RLUU9VOVQ0ODhOQ0tMQ1o0STFQRS4u

Figures 5,6 and 7 shows the questions and the figures 8,9 and 10 shows the authors opinions, my survey consist of 8 questions with 2 types of questions.

Hospital Network Performance and User Experience Survey

This survey is designed to collect feedback from hospital staff regarding their experience with the hospital network system. The questions focus on network performance, reliability, access to the central server, and overall usability during daily tasks. The feedback provided will be used to analyse current network efficiency and identify areas for improvement to better support hospital operations and patient care.

1. How would you rate the overall network performance in your department?

- ◯ Very satisfied
- ◯ Satisfied
- ◯ Neither satisfied nor dissatisfied
- ◯ Dissatisfied
- ◯ Very dissatisfied

2. Do you experience slow network speeds during peak working hours?

- ◯ Yes
- ◯ No
- ◯ Maybe

3. How reliable is access to the central hospital server?

- ◯ Very reliable
- ◯ Somewhat reliable
- ◯ Neither reliable nor unreliable
- ◯ Somewhat unreliable
- ◯ Very unreliable

Figure 5

4. Does the network support your daily tasks without interruptions?

- ◯ Always
- ◯ Most of the time
- ◯ Sometimes
- ◯ Rarely

5. Have you experienced any connection dropouts while using hospital systems?

- ◯ Never
- ◯ Rarely
- ◯ Sometimes
- ◯ Often

6. How satisfied are you with the response time when accessing patient records from 10?

1	2	3	4	5	6	7	8	9	10

Poor — Excellent

7. Do you feel the network is secure when accessing sensitive patient data?

- ◯ Yes
- ◯ No
- ◯ Maybe

Figure 6

8. Which area of the network do you think needs improvement the most?

- ◯ Speed
- ◯ Reliability
- ◯ Security
- ◯ Access to server
- ◯ No improvement needed

Figure 7

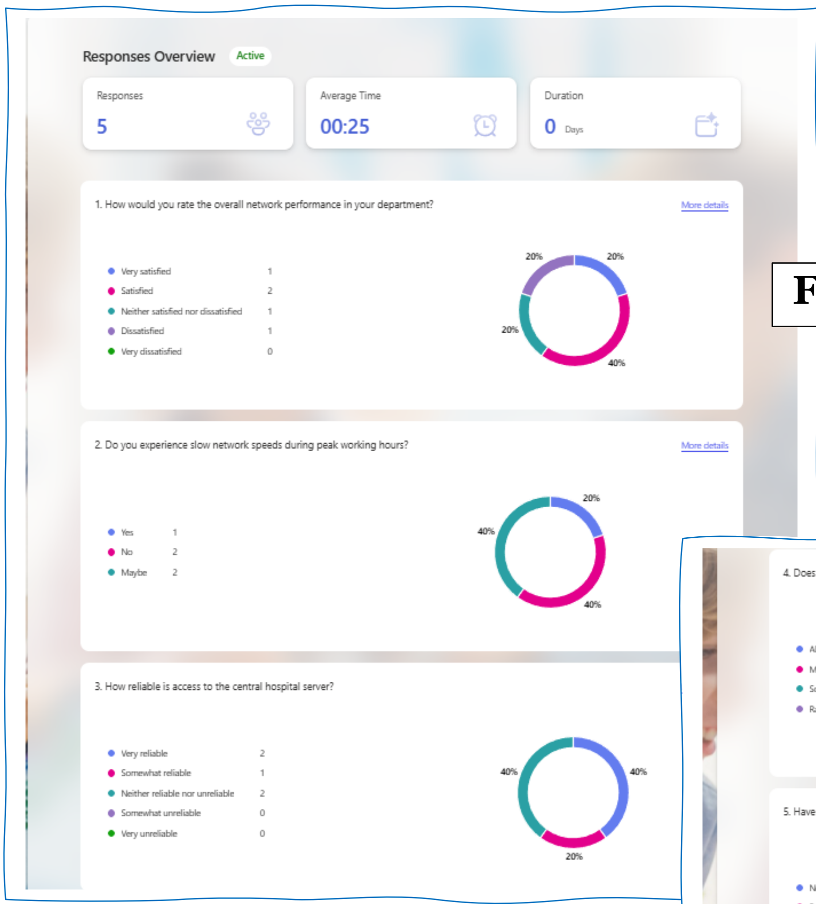


Figure 8

Figure 9

Figure 10

Design Improvements based on user feedback?

The feedback collected from hospital staff shows that the network is generally **performing well**, but there are still some areas that can be improved, Most users rated the overall network performance as satisfied or very satisfied, which indicates that the basic design and VLAN separation are working effectively, However, some users reported experiencing slow network speeds during peak working hours and occasional connection dropouts, This suggests that bandwidth usage may increase when many users access the network at the same time, especially when accessing patient records from the central server. Although server access reliability was rated positively overall, a few users selected neutral responses, showing that performance is not always consistent. All users confirmed that they feel the network is secure when accessing sensitive patient data, which shows that the VLAN based design has successfully improved security. Based on this feedback, improvements focusing on bandwidth management, performance monitoring, and server traffic optimisation were identified to further enhance network efficiency.

Feedback Area	User Feedback Observed	Identified Issue	Design Improvement Applied / Planned
Network Performance	Some users experienced slow speeds during peak hours	Possible bandwidth congestion	Plan to monitor bandwidth usage and prioritise clinical traffic
Server Access	Mostly reliable, but some neutral responses	Inconsistent response times	Server traffic optimisation and regular performance monitoring
Connection Stability	Occasional dropouts reported by a few users	Temporary connection instability	Regular switch and cable health checks
Network Security	All users confirmed the network feels secure	No major security concerns	Maintain VLAN separation and regular security updates
Overall User Experience	Majority satisfied, but improvements suggested	Minor efficiency gaps	Continuous maintenance and future scalability planning

Bringing the Hospital Network Design into "Implementation"

Practical implementation of the hospital network is now carried out using **Cisco Packet Tracer** based on the proposed design. At this stage, I am configuring the network to make sure it works the same way it was planned earlier. This includes setting up VLANs for each floor, enabling automatic IP addressing using DHCP, allowing communication between different floors, and configuring the central server for internal hospital services. The goal of this implementation is to ensure that all departments can access the network smoothly, securely, and without interruption.

What is Cisco Packet Tracer?

Cisco Packet Tracer is a network simulation **tool** that is used to design, build, and test computer networks in a virtual environment. It allows me to create network topologies using routers, switches, servers, and end devices without needing real hardware. Packet Tracer also makes it easier to understand how data flows through a network by visually showing connections and device interactions.In this implementation, Cisco Packet Tracer is used to simulate the hospital network and apply the required configurations such as VLANs, DHCP, routing, and server services. Using Packet Tracer helps in testing the network functionality, identifying possible issues, and confirming that the designed network works correctly before real deployment.

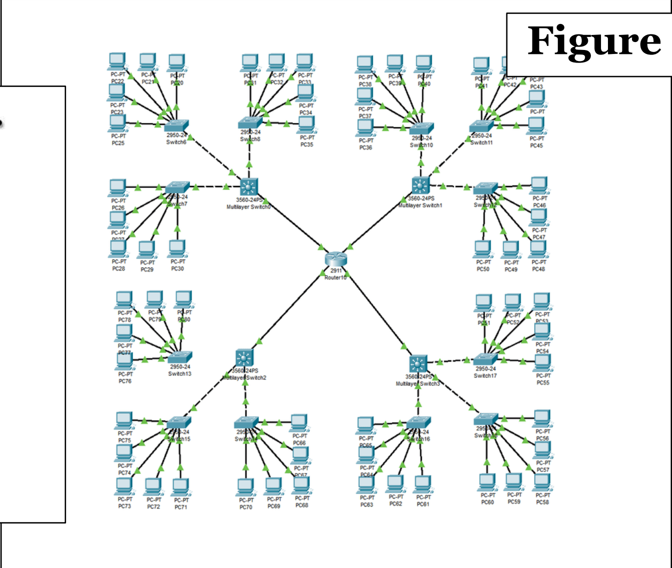


Figure 11

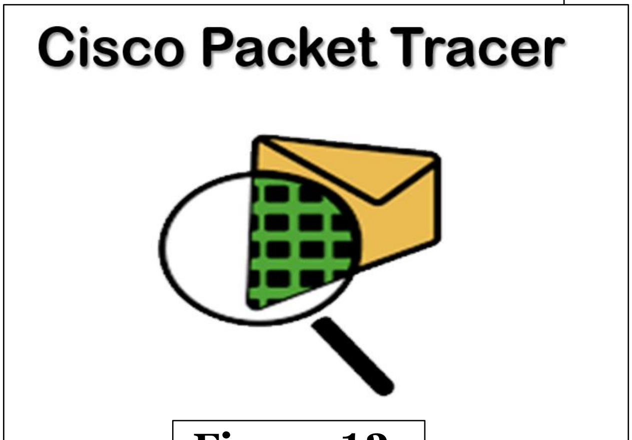


Figure 12

(Cisco Networking Academy, n.d.)

Design Overview previously introduced

The hospital network is designed to support three floors, each with different operational requirements. The network follows an extended star topology where all floor switches are connected to a central core switch, and a router is used to manage communication between different VLANs. This structure makes the network easy to manage and reduces the impact of faults to a single floor.

Each floor is assigned its own VLAN and IP subnet to separate traffic and improve security. The ground floor is used for reception and administrative staff, the first floor hosts doctors' offices and the central server room, and the second floor supports nursing stations and treatment units. A central server is placed on the first floor to provide internal hospital services, This design ensures stable connectivity, secure communication, and supports future expansion of the hospital network.

- **IP addressing/Subnetting/Vlan implementation?**

Aspect	Description	Benefit to Hospital Network
Network Segmentation	Different VLANs and IP ranges are assigned to each floor	Improves security and isolates network traffic between departments
Resource Management	Each floor uses a distinct IP address range	Makes device management and troubleshooting easier
Scalability	Subnetting allows new devices or VLANs to be added	Supports future growth without disrupting existing services
Traffic Optimisation	Devices within the same subnet communicate locally	Reduces unnecessary network traffic and improves performance
Ease of Maintenance	Documented IP addressing and subnet masks	Helps administrators quickly identify and resolve issues
Security Enhancement	VLAN separation limits access between network segments	Minimises security risks and unauthorised access
Effective Communication	Proper IP configuration ensures smooth data transfer	Supports reliable communication within and between floors

(Cisco Networking Academy, n.d.)

IP Addresses of the Hospital Network System?

In the hospital networking project carried out using Cisco Packet Tracer, IP addresses and default gateways are assigned based on the network design requirements. Each floor of the hospital is placed in a separate VLAN with its own IP subnet to ensure proper network segmentation, security, and efficient communication. The addressing scheme is planned in a structured way to support daily hospital operations and allow future expansion.

Device / Group	IPv4 Address / Range	Subnet Mask	Default Gateway
Hospital Server	192.168.20.5	255.255.255.0	192.168.20.1
Ground Floor PCs	192.168.10.10 to 192.168.10.14	255.255.255.0	192.168.10.1
Ground Floor Printers	192.168.10.50 to 192.168.10.53	255.255.255.0	192.168.10.1
First Floor PCs and Laptops	192.168.20.10 to 192.168.20.21	255.255.255.0	192.168.20.1
Second Floor PCs and Laptops	192.168.30.10 to 192.168.30.22	255.255.255.0	192.168.30.1

Each floor in the hospital is assigned a unique VLAN and IP subnet to improve network organisation and security. The ground floor uses the 192.168.10.0/24 network for administrative systems and printers, while the first floor uses the 192.168.20.0/24 network for doctors' systems and the central hospital server. The second floor uses the 192.168.30.0/24 network to support nursing stations and treatment units.

The server is assigned a fixed IP address to ensure reliable access to patient records and hospital services. Default gateways are configured for each VLAN to allow controlled communication between different floors through the router. This IP addressing scheme supports efficient communication, easier troubleshooting, and future scalability of the hospital network.

Subnet Mask Calculation of the Hospital Network?

Floor / VLAN	IP Network	Hosts Required	Subnet Mask	CIDR	Usable Hosts per Subnet	IP Address Range
Ground Floor (VLAN 10)	192.168.10.0	80 devices	255.255.255.0	/24	254 hosts	192.168.10.1 to 192.168.10.254
First Floor (VLAN 20)	192.168.20.0	60 devices	255.255.255.0	/24	254 hosts	192.168.20.1 to 192.168.20.254
Second Floor (VLAN 30)	192.168.30.0	40 devices	255.255.255.0	/24	254 hosts	192.168.30.1 to 192.168.30.254

Floor / VLAN	Subnet Mask Calculation and Usage
Ground Floor (VLAN 10)	The Ground Floor uses the subnet mask 255.255.255.0 (/24), which provides up to 254 usable IP addresses. This is sufficient to support administrative PCs, reception systems, printers, and future device expansion.
First Floor (VLAN 20)	The First Floor is assigned a /24 subnet to support doctors' systems and the central hospital server. This subnet allows reliable communication and enough IP addresses for current and future medical systems.
Second Floor (VLAN 30)	The Second Floor also uses a /24 subnet, ensuring enough IP addresses for nursing stations and treatment units. This helps maintain stable connectivity during daily hospital operations.
Overall Network Design	Using the same subnet mask across all floors simplifies network management, improves consistency, and allows easier troubleshooting and scalability.

Installation and Configuration in PT?

Device **installation** refers to the process of adding network devices into the Packet Tracer workspace. During this stage, devices such as routers, switches, PCs, and servers are placed into the logical network diagram. Installation focuses only on the physical and visual layout of the network and does not involve any configuration. At this point, devices are not operational and cannot communicate until they are configured.

Device **configuration** is the process of setting up the installed devices so that they can function within the network. This includes assigning IP addresses, configuring VLANs, enabling routing, and setting up services such as DHCP and web access. Configuration makes the devices operational and allows them to perform their intended roles within the hospital network.

Aspect	Device Installation	Device Configuration
Definition	The process of adding network devices into the Packet Tracer workspace	The process of setting up device parameters so they can function in the network
Main Focus	Physical and visual layout of the network	Functional and operational setup of devices
Actions Performed	Placing routers, switches, PCs, and servers in the workspace	Assigning IP addresses, configuring VLANs, routing, and services
Network Functionality	Devices are not operational and cannot communicate	Devices become fully operational and able to communicate
Tools Used	Drag-and-drop interface in Packet Tracer	CLI or graphical configuration interface
Stage in Implementation	Initial stage of network setup	Final stage that enables network communication
Example from Hospital Network	Placing core switch, floor switches, router, PCs, and server	Setting VLANs, enabling DHCP, inter-VLAN routing, and server services

Server Configuration and FTP

The hospital server is configured first because it is a critical component of the network and is used to provide internal services to hospital staff. The server is placed on the first floor and assigned a static IP address to ensure reliable and consistent access. Assigning a static IP prevents service disruption and makes it easier for users from different floors to access hospital resources.

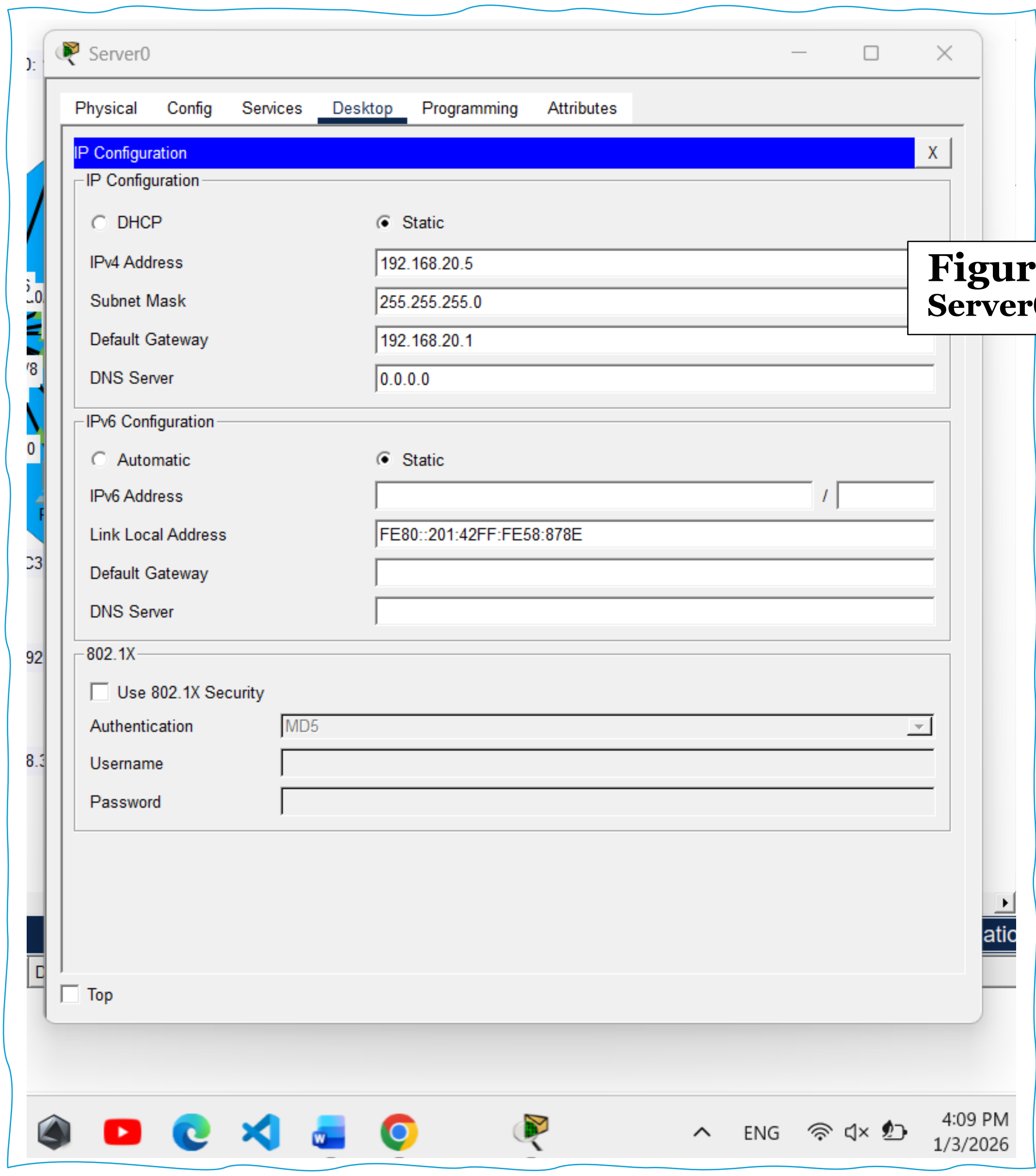


Figure 13: Server0

(Burgess, 2004)

Then Ftp Setup

After configuring the server's IP address, the FTP service is enabled to allow file sharing within the hospital network. FTP is used to transfer files such as internal documents and reports between staff and the central server. Enabling the FTP service ensures that users from different departments can upload and download files securely within the local network.

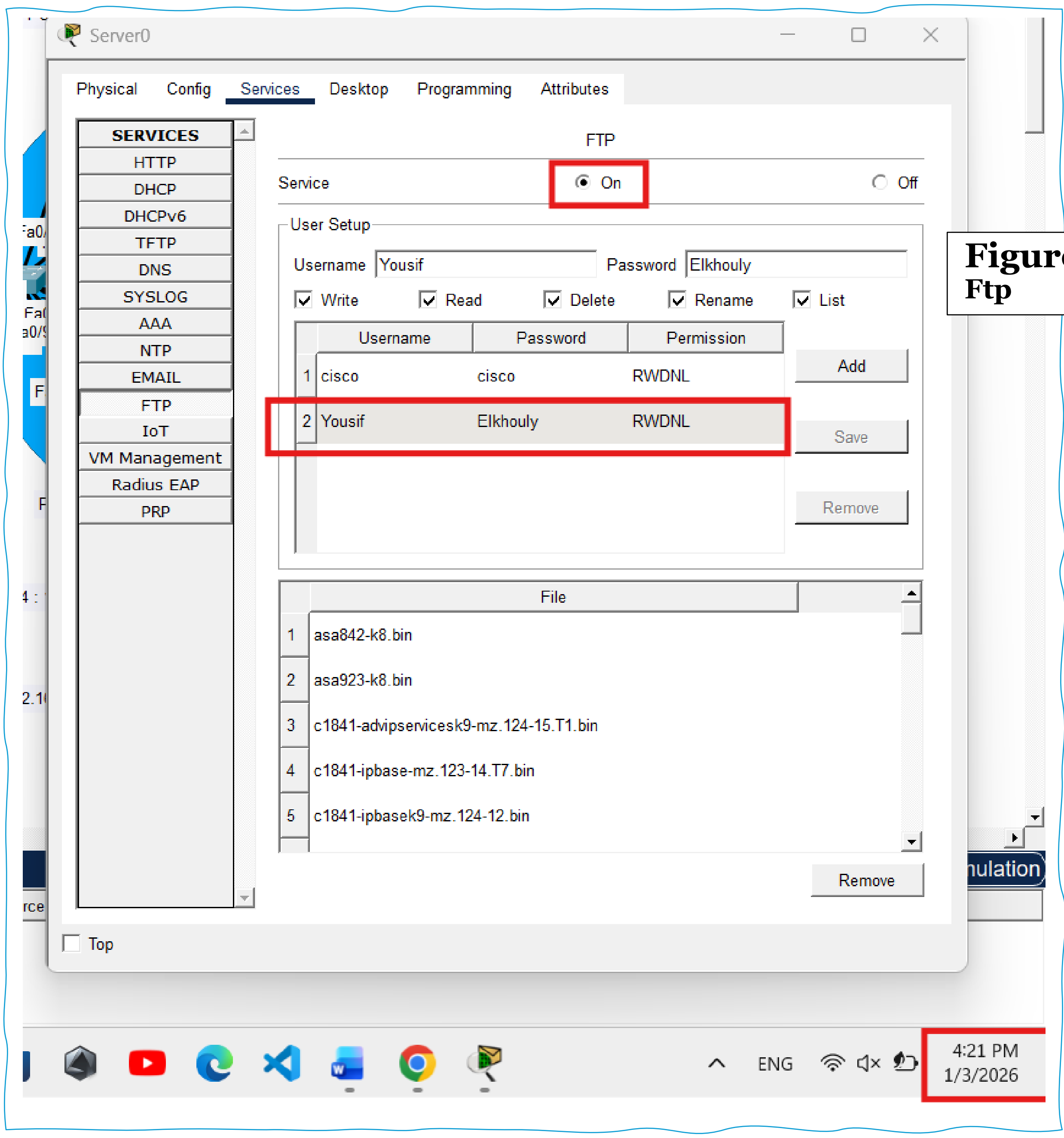


Figure 14: Ftp

(Burgess, 2004)

Then Creating a File for FTP Upload

To test the FTP service, a sample file is created on a client PC. This file is later uploaded to the FTP server to confirm that file transfer functionality is working correctly across the network. Device chosen “PC5”

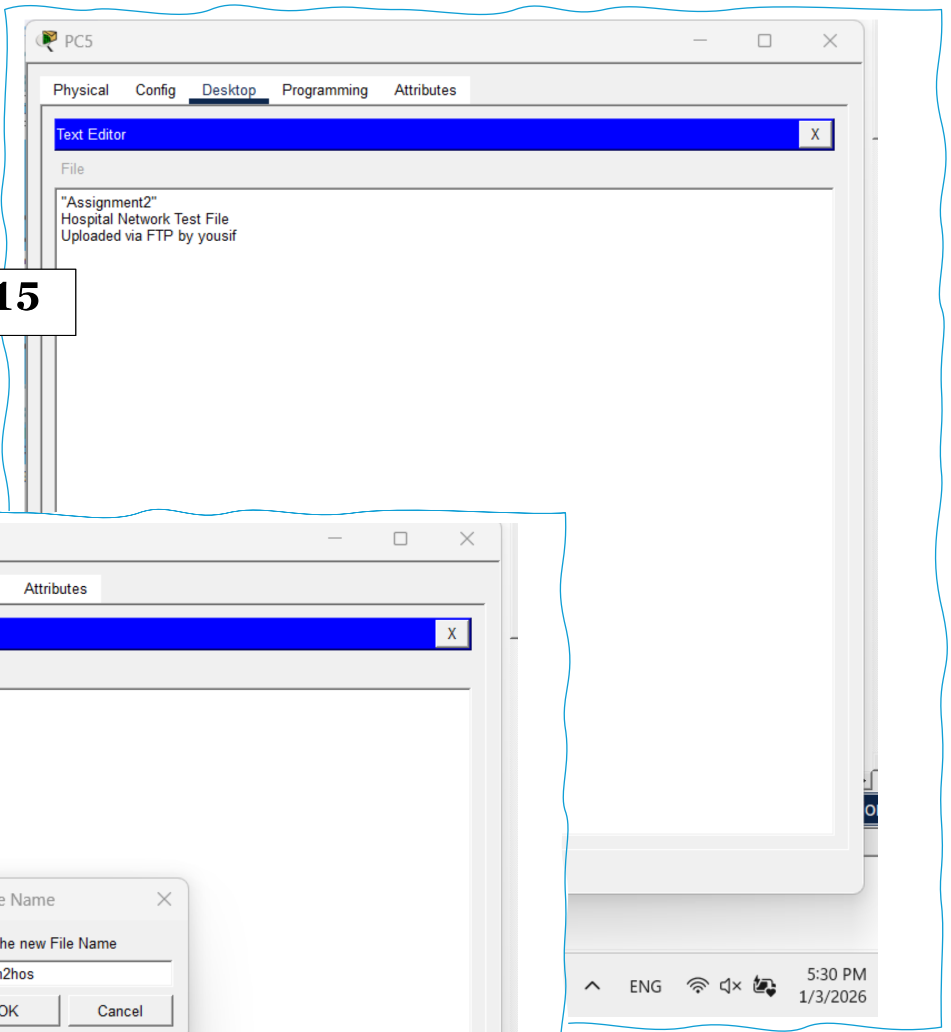


Figure 15

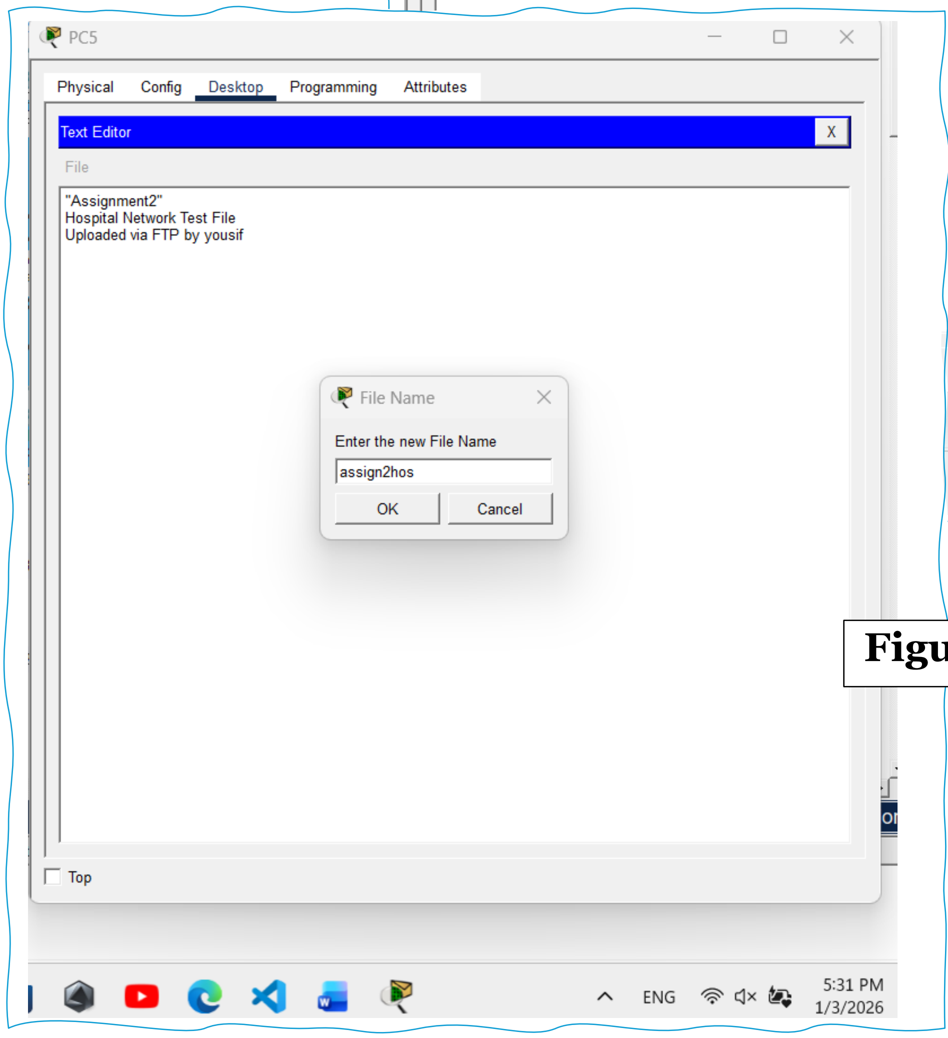


Figure 16

Then Downloading File from FTP Server

To further verify the FTP functionality, the uploaded file is downloaded from the FTP server. Successful download confirms two way file transfer between client devices and the server. “By using the same PC for sure “PC5””

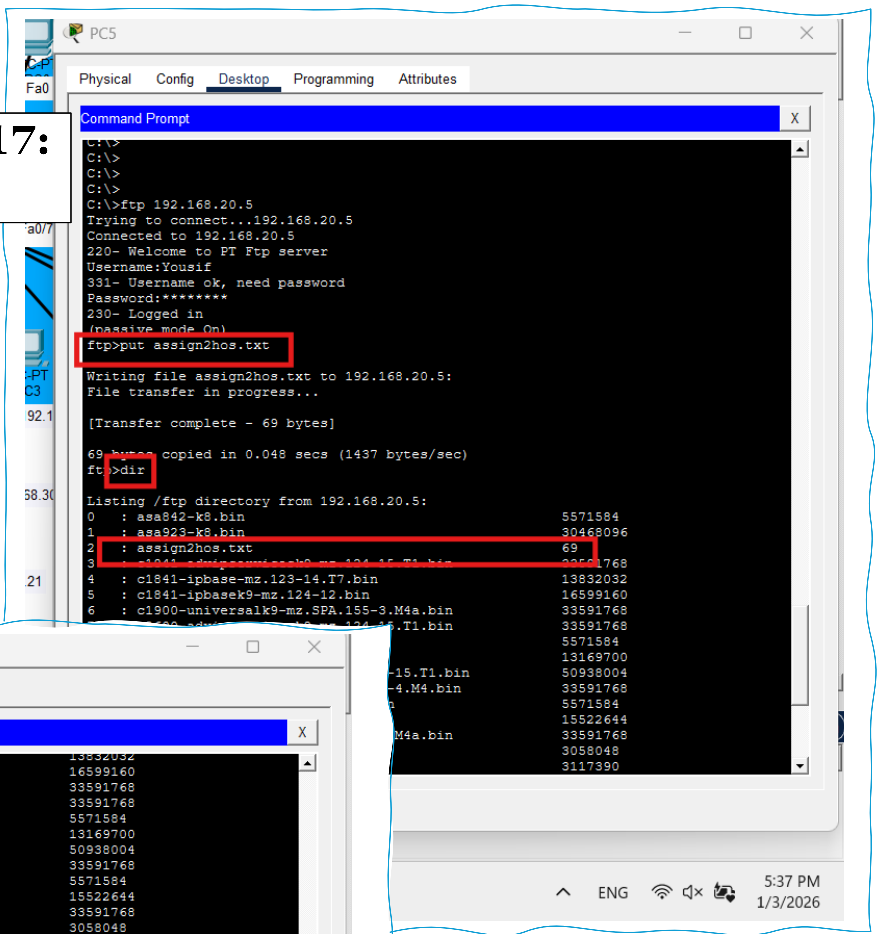


Figure 17: put

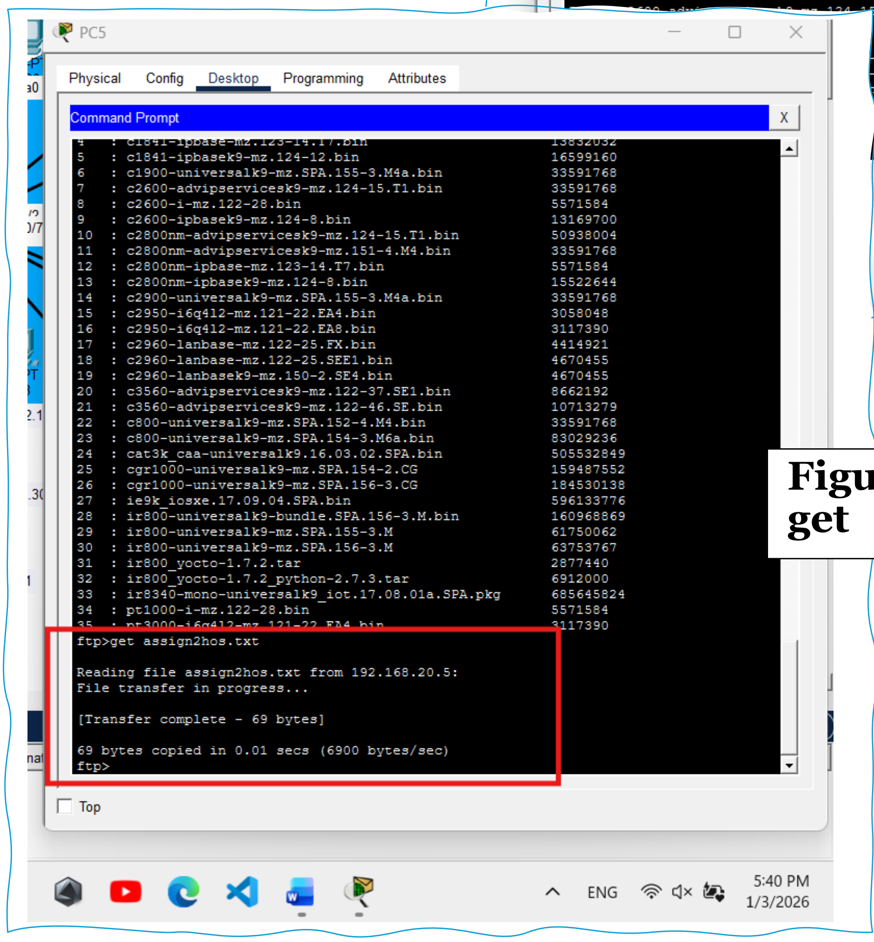


Figure 18: get

Virtual Local Area Network “Vlan”

Virtual Local Area Networks “VLANs” are used to logically **divide** a physical network into smaller, more manageable segments, Instead of placing all devices in one large network, VLANs allow devices to be grouped based on their function or location. This improves network performance by reducing broadcast traffic and enhances security by isolating sensitive systems from general access.

Actually in the hospital network, VLANs are implemented to **separate traffic between different floors** and departments. Each floor is assigned its own VLAN, which ensures that administrative systems, doctors’ workstations, and nursing units operate within their own network segments, This approach helps protect sensitive patient data, improves overall network efficiency, and makes the network easier to manage and scale as the hospital grows. ***(Donahue, 2011)***

Benefits?

Aspect	Description
Logical Segmentation	VLANs divide a single physical network into multiple logical broadcast domains, improving organisation and control of network traffic.
Flexibility and Scalability	Devices can be grouped based on department or function rather than physical location, making it easier to add new devices and scale the network.
Broadcast Control	Broadcast traffic is limited within each VLAN, reducing unnecessary traffic and improving overall network performance.
Security	VLANs isolate sensitive systems into separate network segments, reducing the risk of unauthorised access and improving data protection.
Inter-VLAN Routing	Communication between different VLANs is enabled through a router or Layer 3 device while maintaining logical separation.
Management and Troubleshooting	VLAN-based segmentation simplifies network management and helps administrators quickly identify and resolve issues.

How to Implement VLAN?

Since VLAN implementation is the process of configuring switches to logically separate devices into different network segments, In this hospital network, VLANs are created to group devices based on their floor and function rather than their physical connection. This helps improve security, reduce broadcast traffic, and make the network easier to manage.

So to implement VLANs, VLAN IDs are first created on the switches. After creating the VLANs, switch ports connected to end devices are assigned to the appropriate VLAN based on their floor. Trunk links are then configured between switches to allow traffic from multiple VLANs to pass through a single connection. This ensures that all VLANs can communicate with the router for inter-VLAN routing while remaining logically separated within the network.

The purpose of Vlan?

Purpose	Explanation
Improved Network Performance	VLANs improve network performance by dividing a large network into smaller broadcast domains. This reduces unnecessary broadcast traffic and ensures that data packets stay within their relevant VLAN, helping optimise network resources and reduce congestion.
Enhanced Security	VLANs increase network security by isolating devices based on their roles or departments. Sensitive systems, such as medical or administrative devices, are separated from general access, reducing the risk of unauthorised access and improving data protection.
Ease of Administration	VLANs simplify network management by allowing devices to be grouped logically rather than physically. This makes it easier for administrators to manage users, apply policies, troubleshoot issues, and adapt the network to organisational changes without rewiring.

Vlan types?

VLANs can be implemented in different ways depending on how network devices are grouped and managed. The two most commonly used types of VLANs are port-based (static) VLANs and user-based (dynamic) VLANs. Each type has its own advantages and is suitable for different network environments. In this hospital network, the VLAN type is chosen based on simplicity, control, and ease of management.

VLAN Type	Description	How It Works	Typical Use Case
Port-Based (Static) VLAN	VLAN membership is assigned to specific switch ports	Devices are placed into VLANs based on the physical port they connect to	Suitable for environments where device locations are fixed, such as hospital floors
User-Based (Dynamic) VLAN	VLAN membership is assigned based on user authentication or device identity	VLANs are assigned dynamically using authentication methods such as 802.1X	Suitable for environments with mobile users or frequently changing device locations

VLAN Configuration Using Command Line Interface or CLI?

The VLAN configuration using the Command Line Interface "CLI" is carried out to logically segment the network and assign devices to their respective VLANs. Using the CLI allows precise control over switch configuration, including creating VLANs, naming them for easy identification, assigning switch ports as access ports, and configuring trunk links between switches. This method ensures that devices on different floors are separated into their own VLANs while still allowing controlled communication through trunk ports and routing. The following table explains the key commands used during VLAN implementation and their purpose within the hospital network.

(Donahue, 2011)

Step	Command Used	Command Purpose / Explanation
1	`enable`	Enters privileged EXEC mode to allow access to configuration commands
2	`configure terminal`	Switches to global configuration mode
3	`vlan 10`	Creates VLAN 10 for the Ground Floor
4	`name ground_floor`	Assigns a descriptive name to VLAN 10 for identification
5	`exit`	Exits VLAN configuration mode
6	`vlan 20`	Creates VLAN 20 for the First Floor
7	`name first_floor`	Assigns a name to VLAN 20
8	`exit`	Exits VLAN configuration mode
9	`vlan 30`	Creates VLAN 30 for the Second Floor
10	`name second_floor`	Assigns a name to VLAN 30
11	`exit`	Exits VLAN configuration mode
12	`interface fa0/8`	Enters interface configuration mode for FastEthernet 0/8
13	`switchport mode access`	Configures the interface as an access port
14	`switchport access vlan 10`	Assigns the interface to VLAN 10
15	`exit`	Exits interface configuration mode

16	`interface fa0/1`	Enters interface configuration mode for FastEthernet 0/1
17	`switchport mode access`	Configures the interface as an access port
18	`switchport access vlan 20`	Assigns the interface to VLAN 20
19	`exit`	Exits interface configuration mode
20	`interface fa0/2`	Enters interface configuration mode for FastEthernet 0/2
21	`switchport mode access`	Configures the interface as an access port
22	`switchport access vlan 30`	Assigns the interface to VLAN 30
23	`exit`	Exits interface configuration mode
24	`interface range fa0/8, fa0/1, fa0/2`	Selects multiple interfaces at the same time
25	`switchport mode trunk`	Configures selected interfaces as trunk ports
26	`exit`	Exits interface range configuration mode
27	`exit`	Exits global configuration mode and completes VLAN setup

CLI in my project

Figure 19: Switch0 the core switch

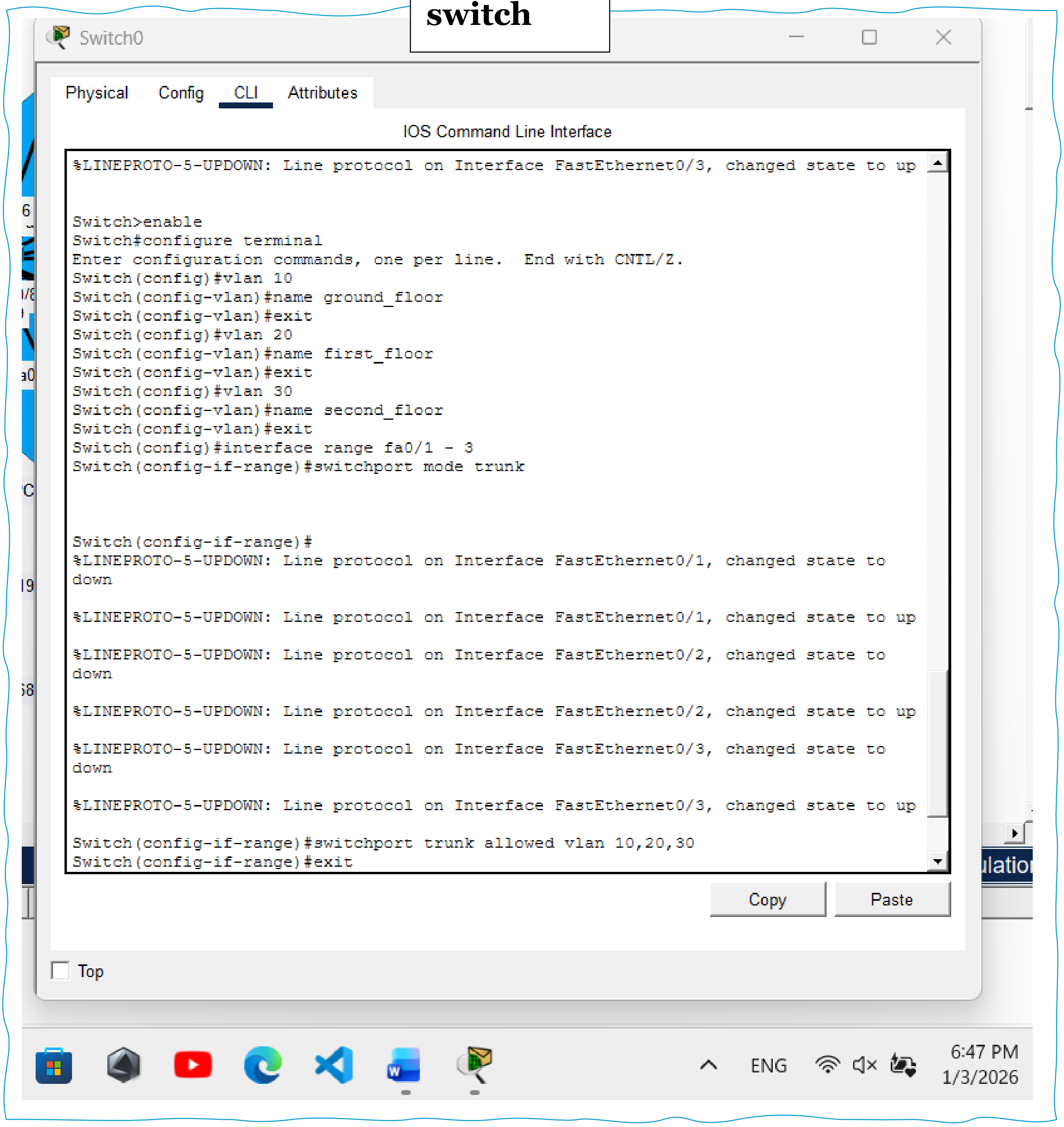


Figure 20:
Switch1 the ground floor switch

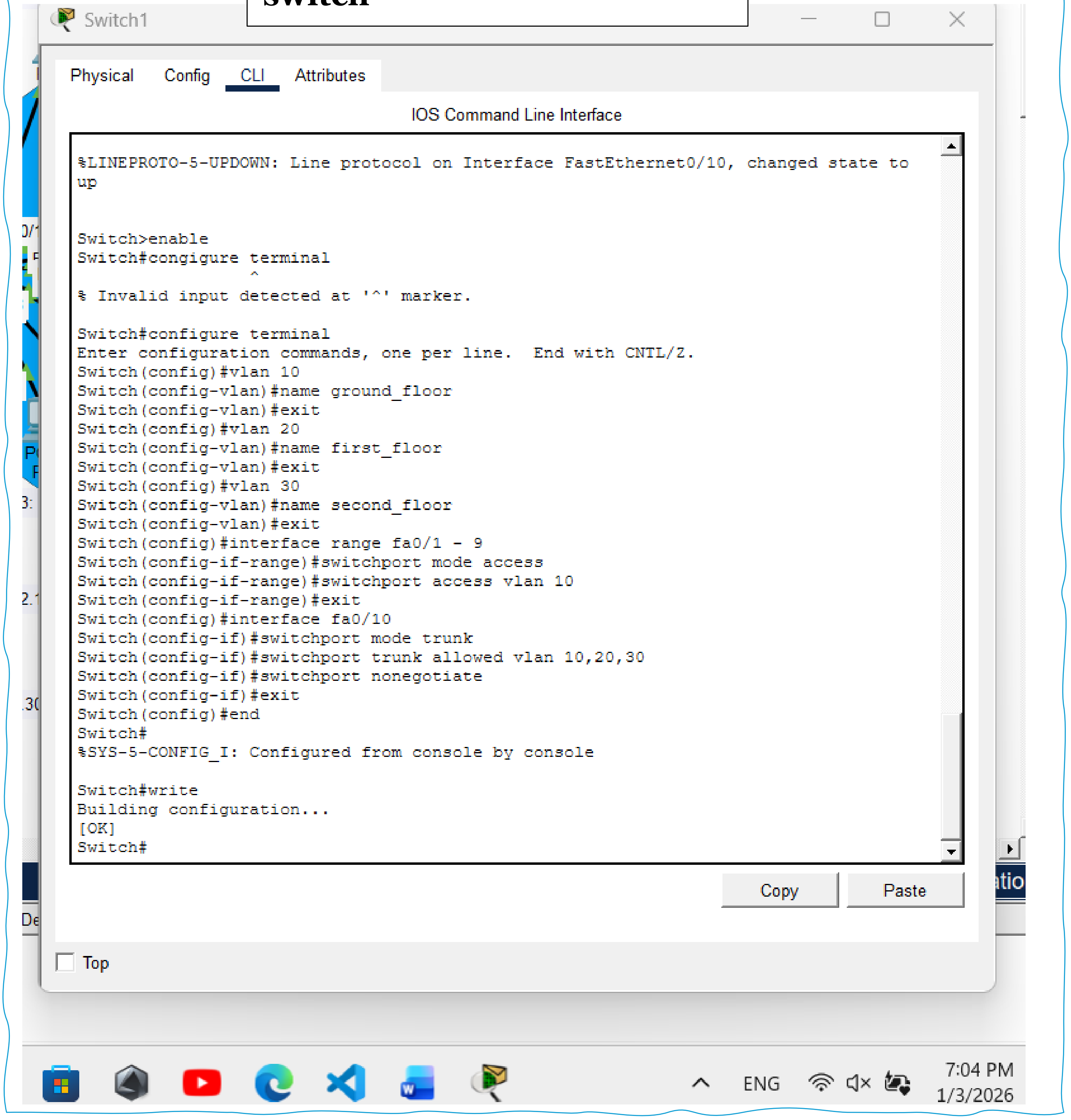


Figure 21:
Switch2 the first floor switch

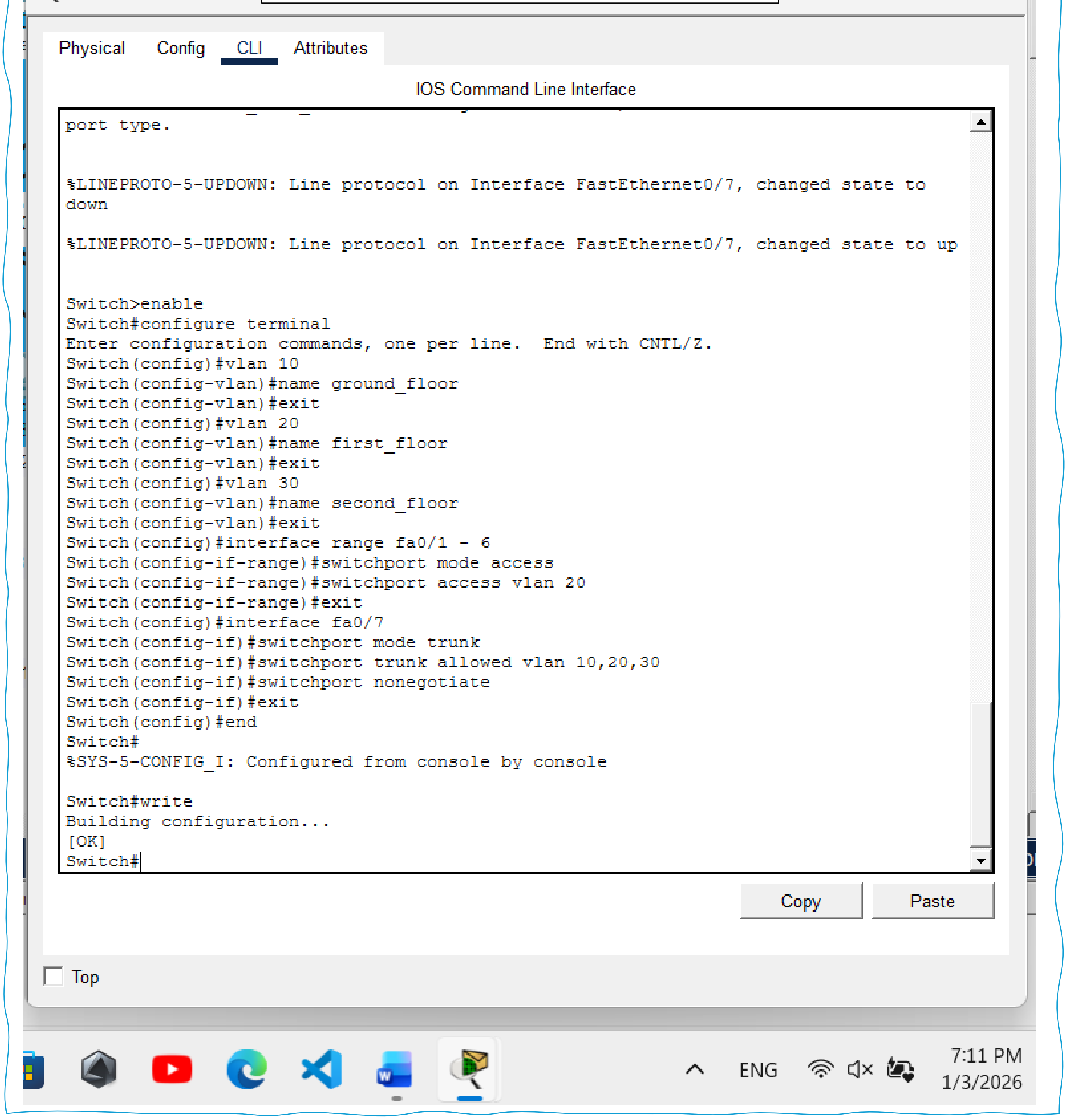
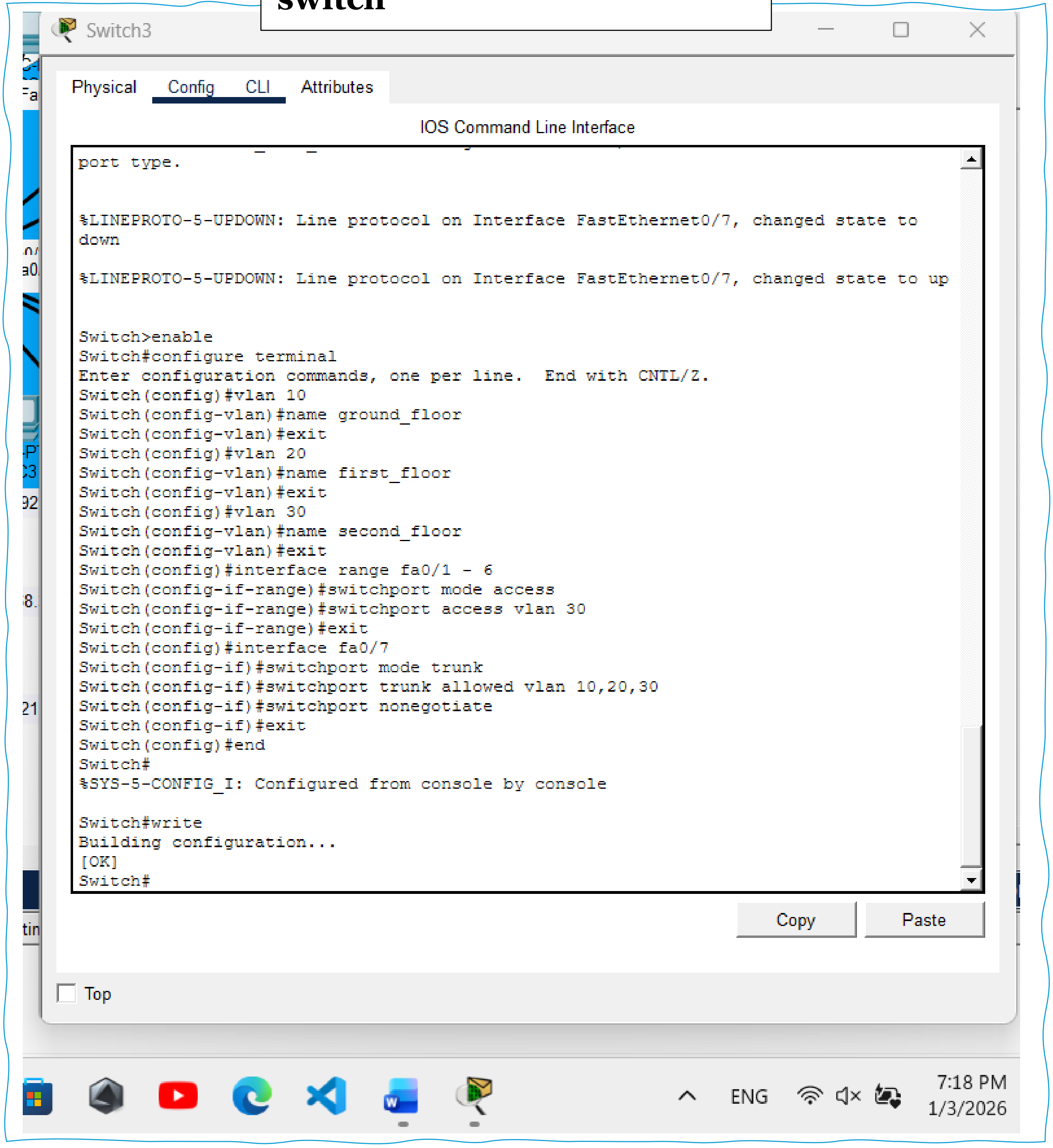


Figure 22:
Switch2 the Second floor switch



Devices Configuration in order “Floor by floor”

1 – Ground floor

On the ground floor, all PCs and printers are part of VLAN 10, which is used for reception and administrative operations. Each device on this floor is configured with an IP address from the 192.168.10.0/24 network and uses 192.168.10.1 as the default gateway. This configuration ensures that all administrative systems can communicate within the same VLAN and access shared resources such as printers efficiently. Assigning IP addresses in an organised manner also helps with easier network management and troubleshooting.

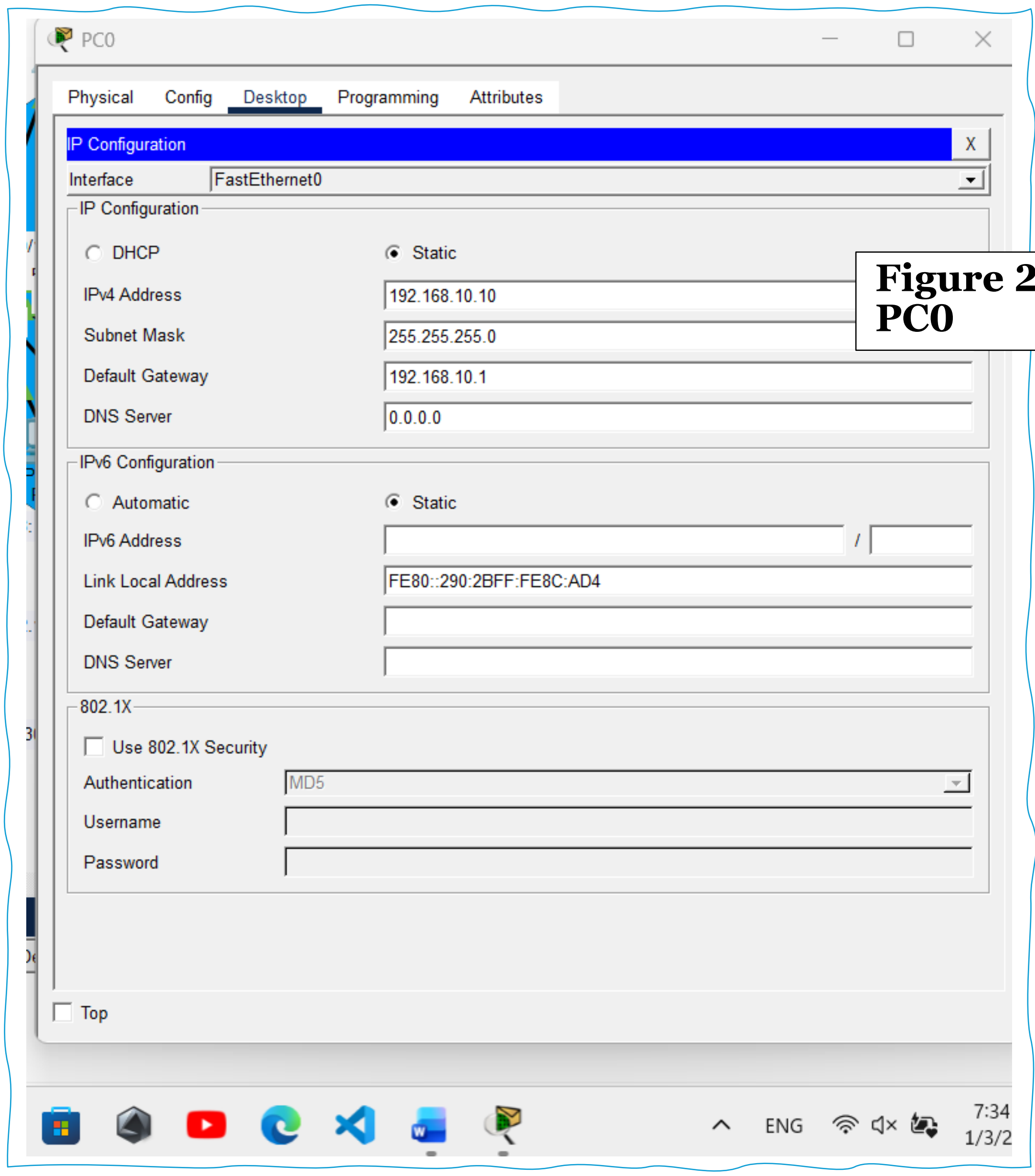


Figure 23: PC0

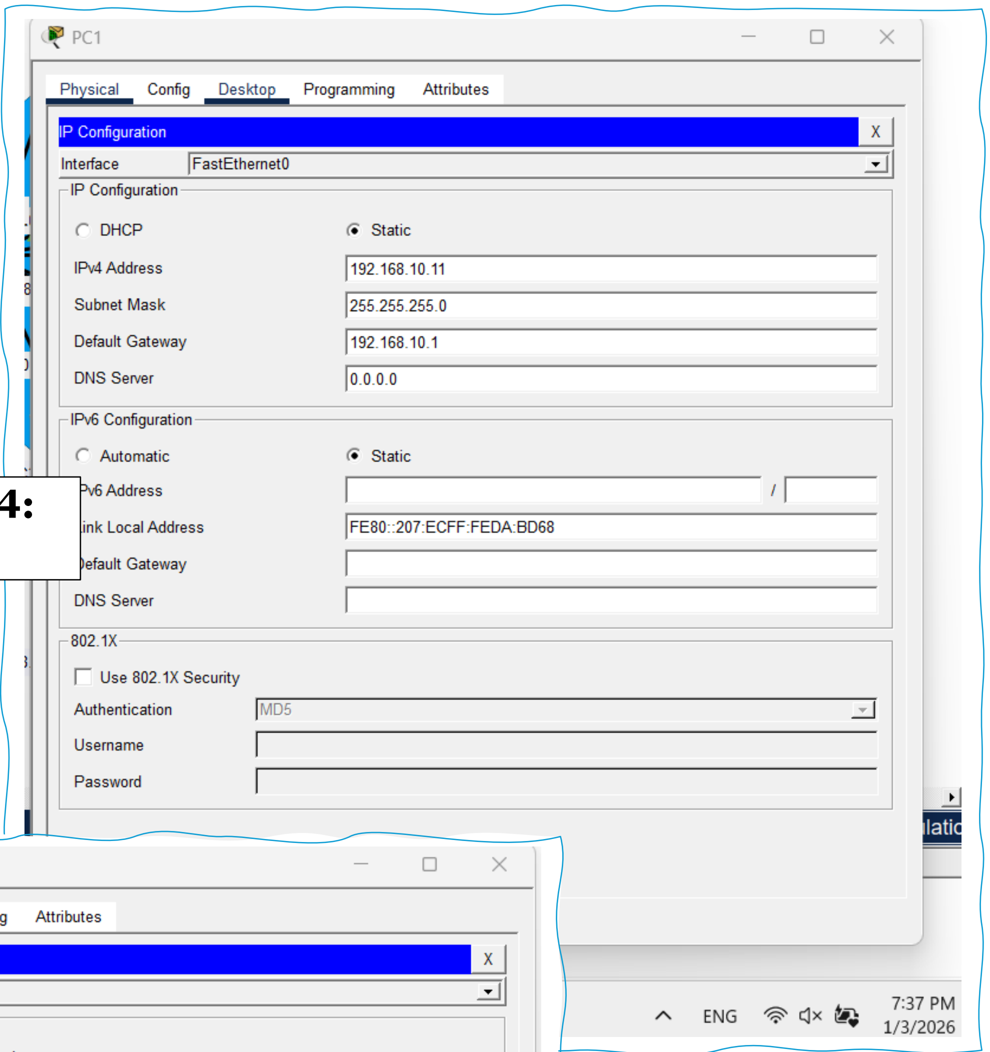


Figure 24: PC1

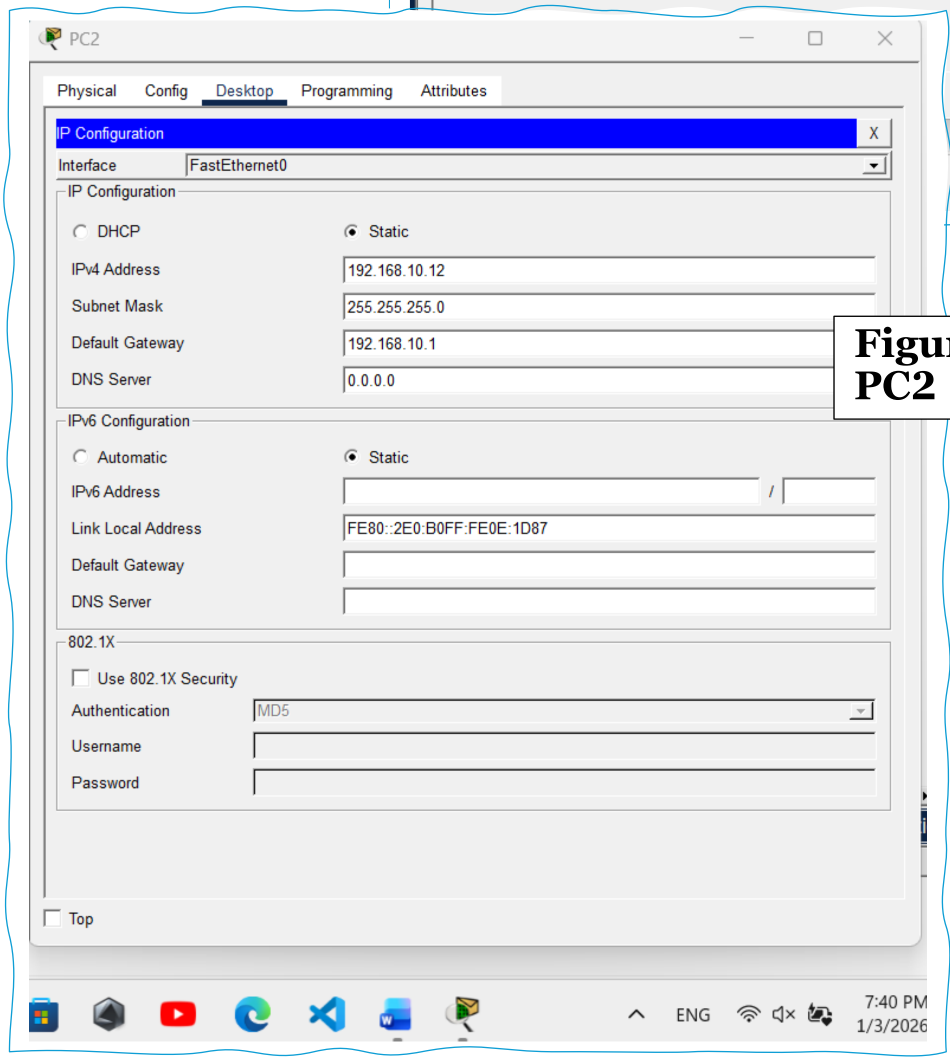
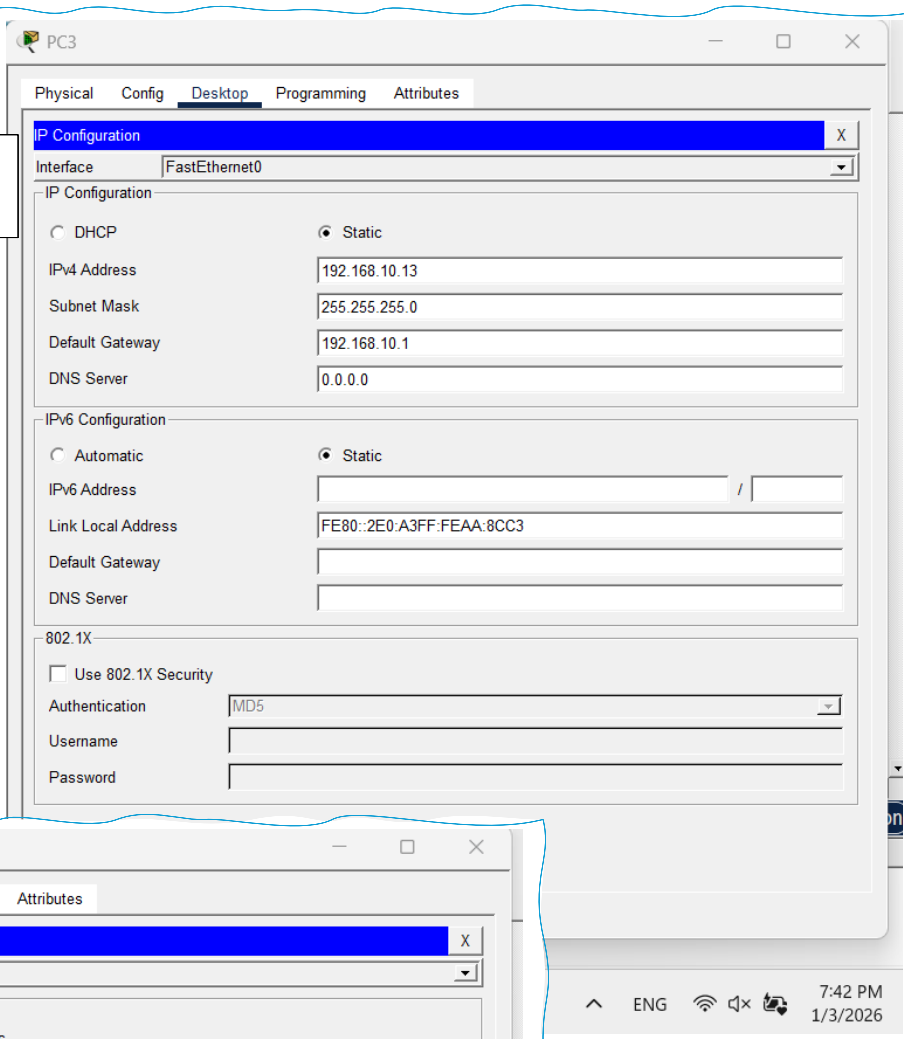


Figure 25: PC2

Figure 26: PC3



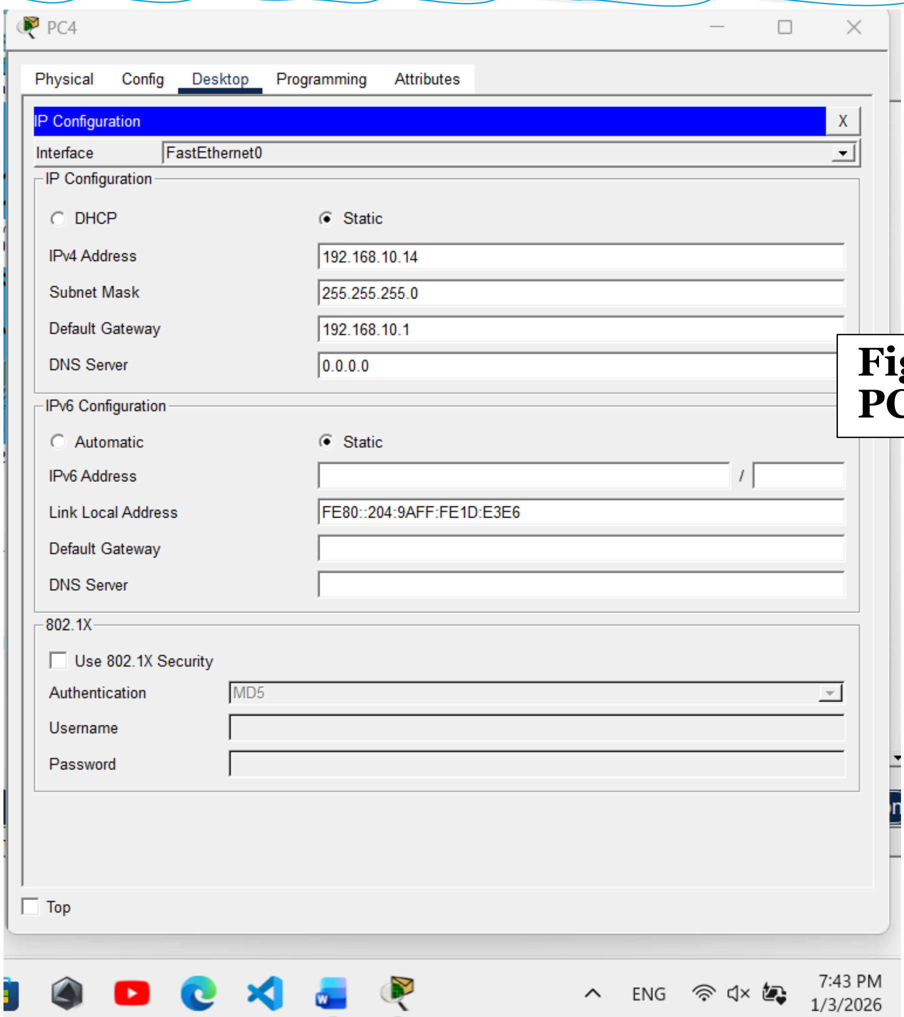


Figure 27: PC4

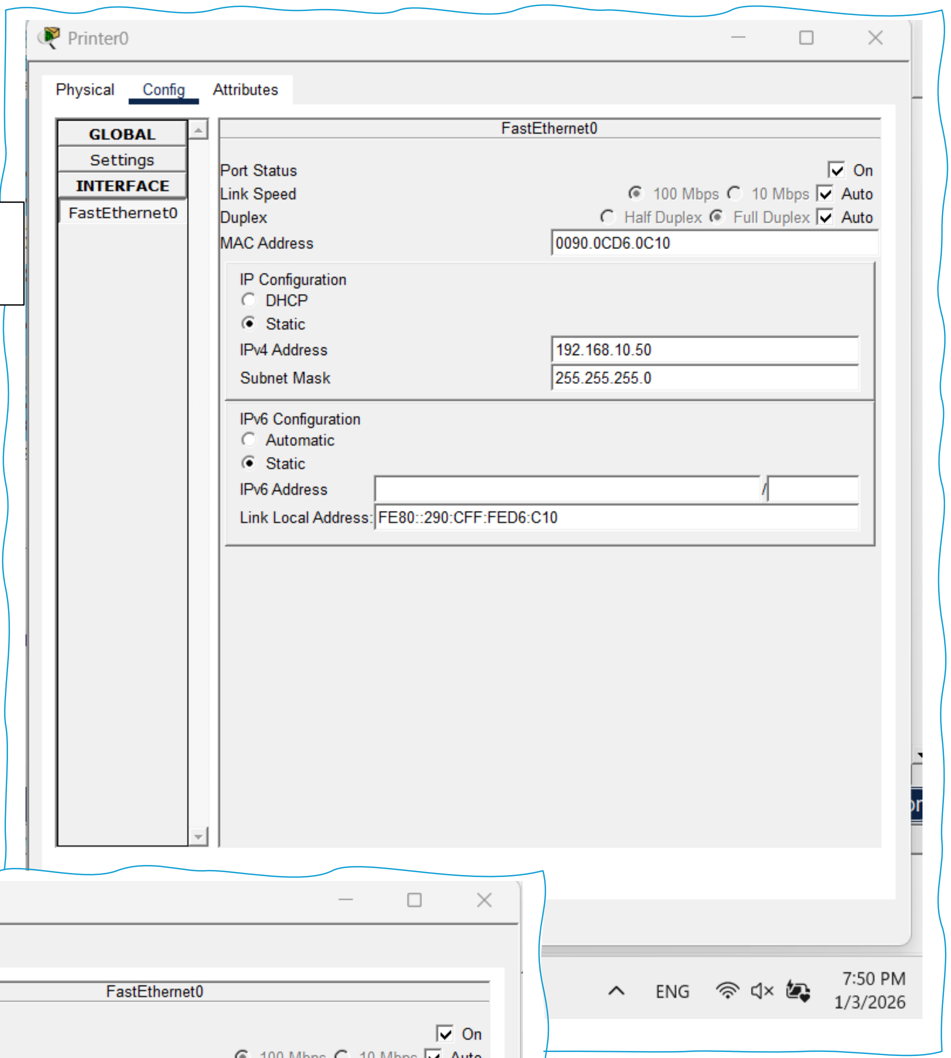


Figure 28: Printer0

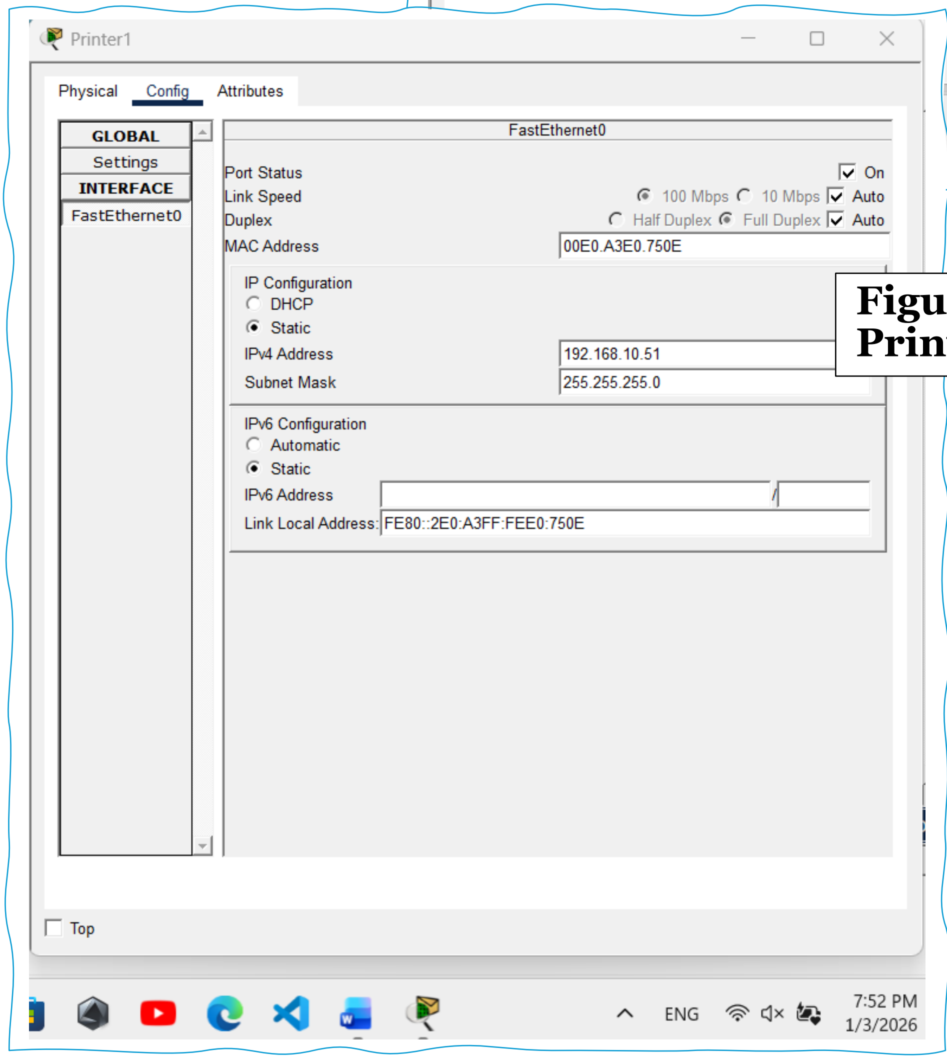
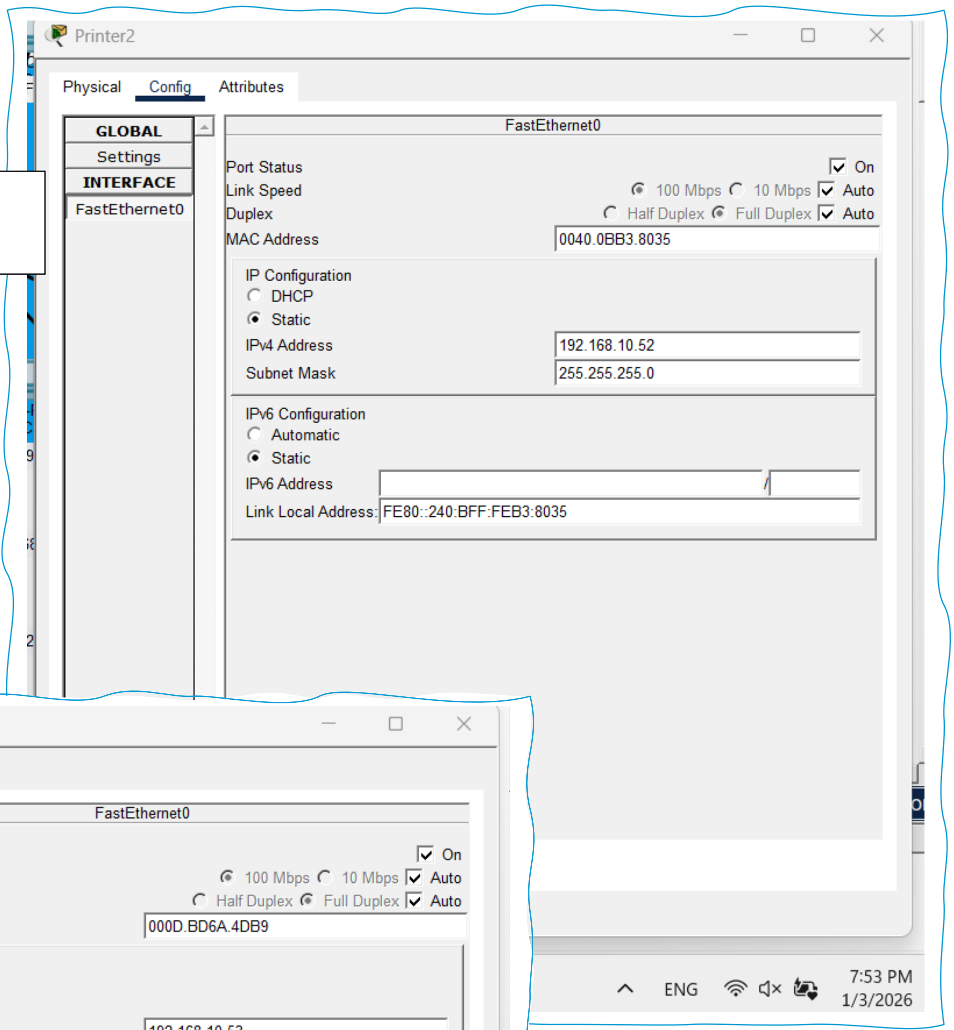


Figure 29: Printer1

Figure 30: Printer2



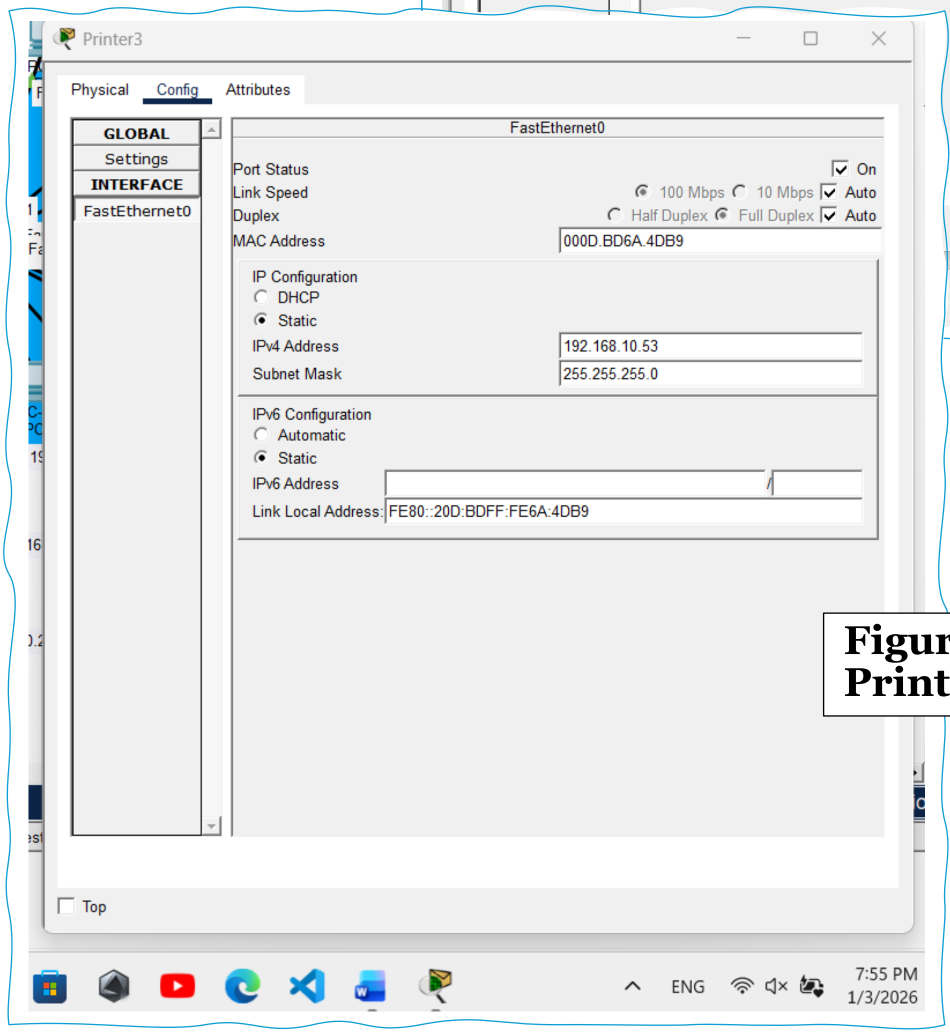


Figure 31: Printer3

2 -first floor

Devices on the second floor, which includes nursing stations and treatment units, are configured under **VLAN 30**. Each PC and laptop is assigned an IP address from the **192.168.30.0/24** network and uses **192.168.30.1** as the default gateway. This configuration supports smooth communication for nursing staff and patient care systems while maintaining network isolation from other floors.

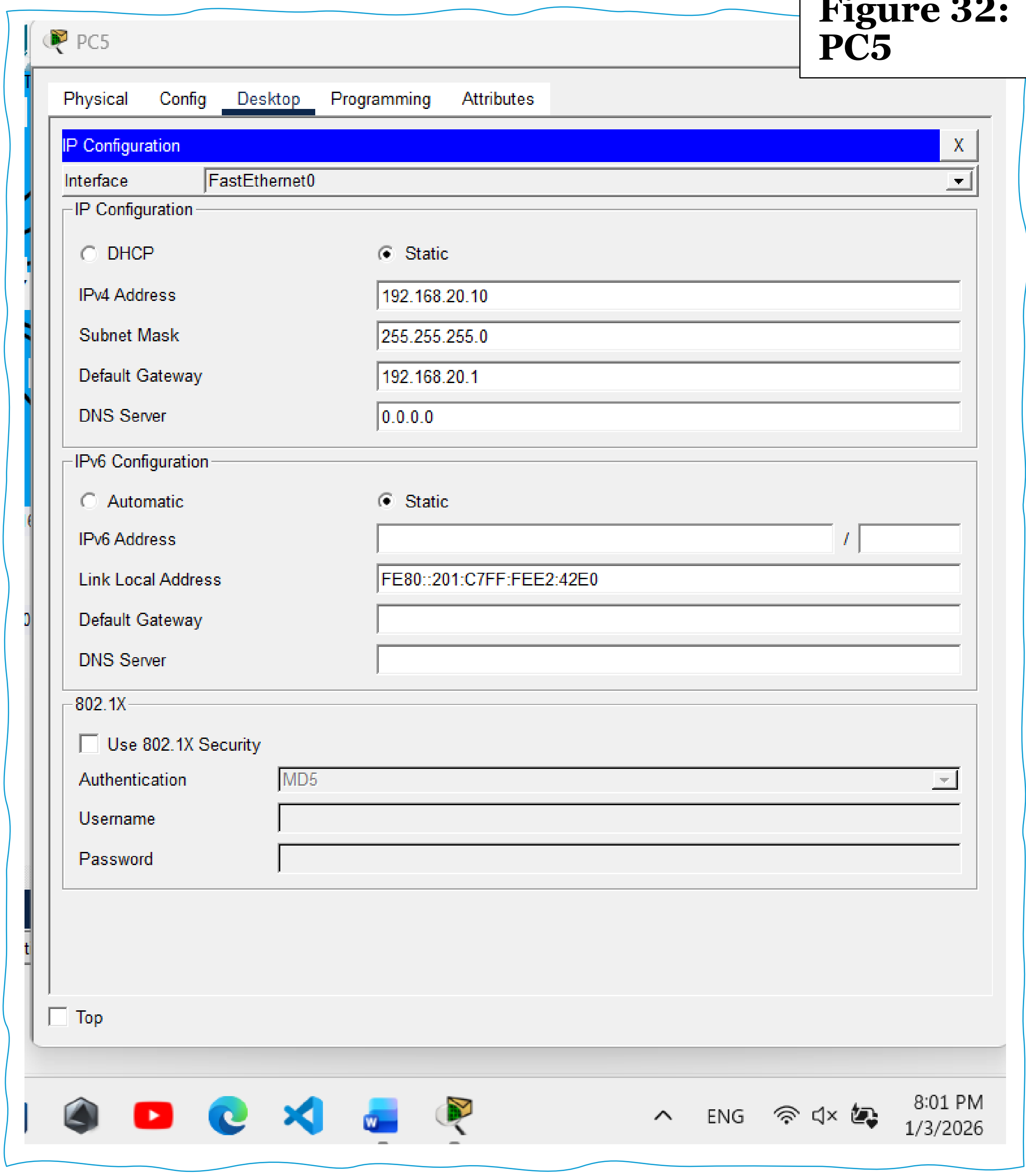
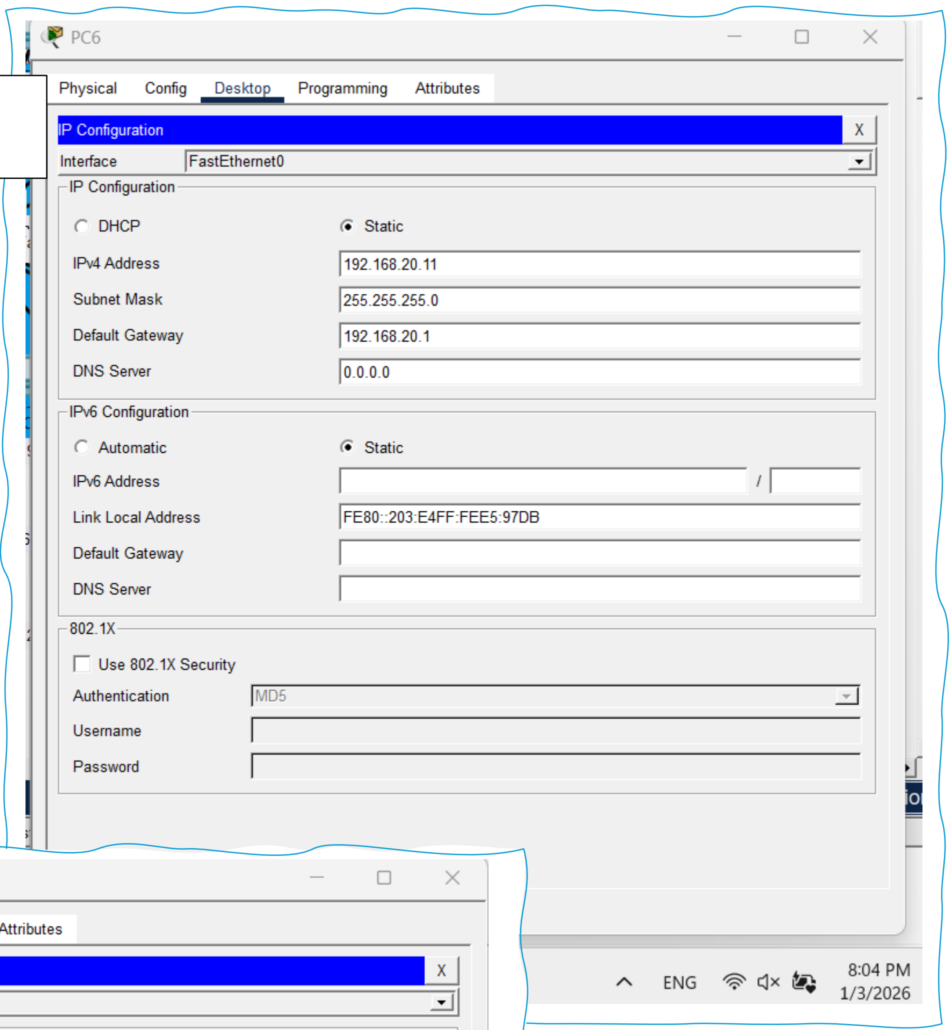


Figure 32: PC5

Figure 33: PC6



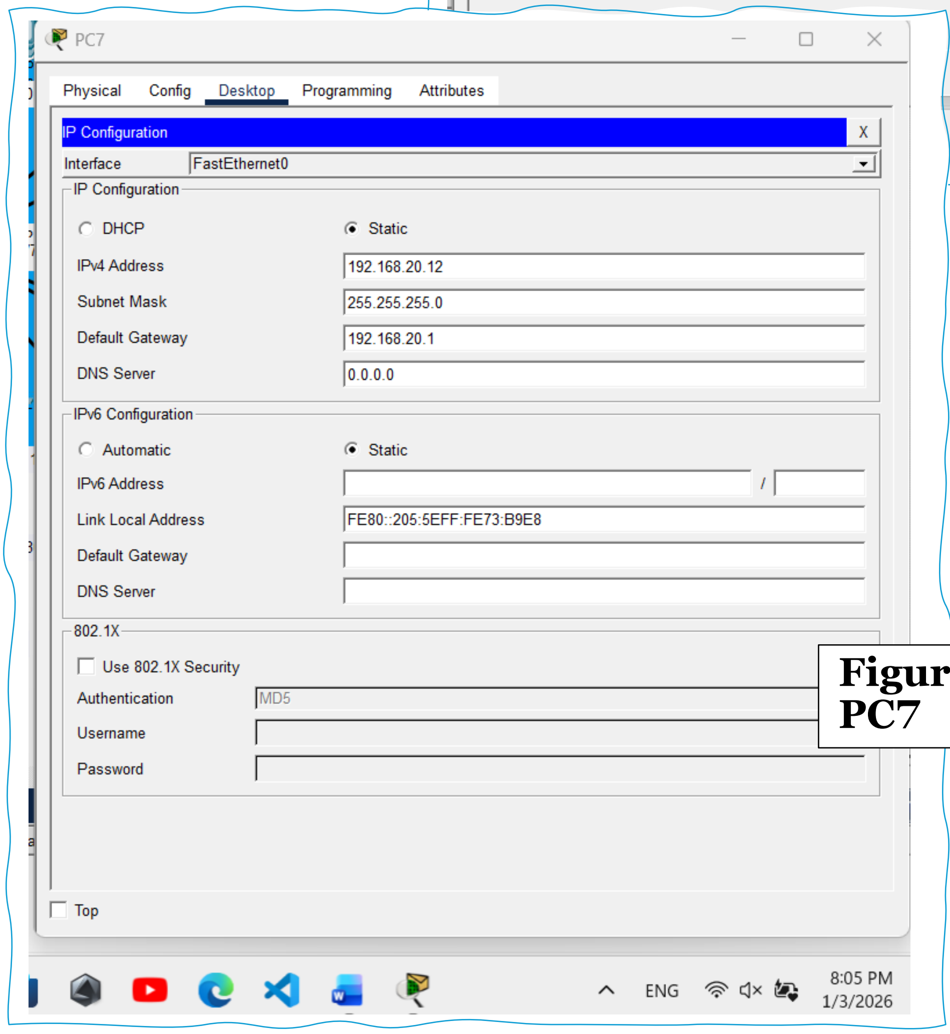
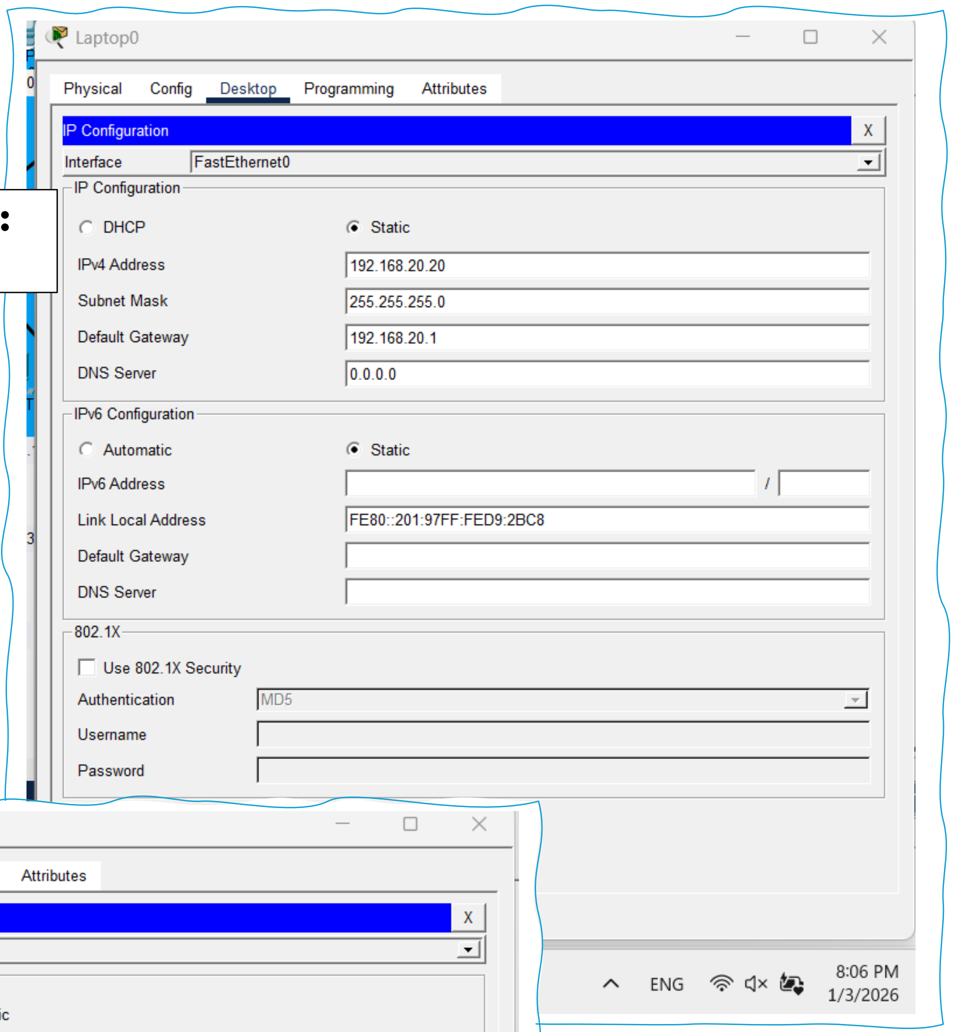


Figure 34: PC7

Figure 35: Laptop0



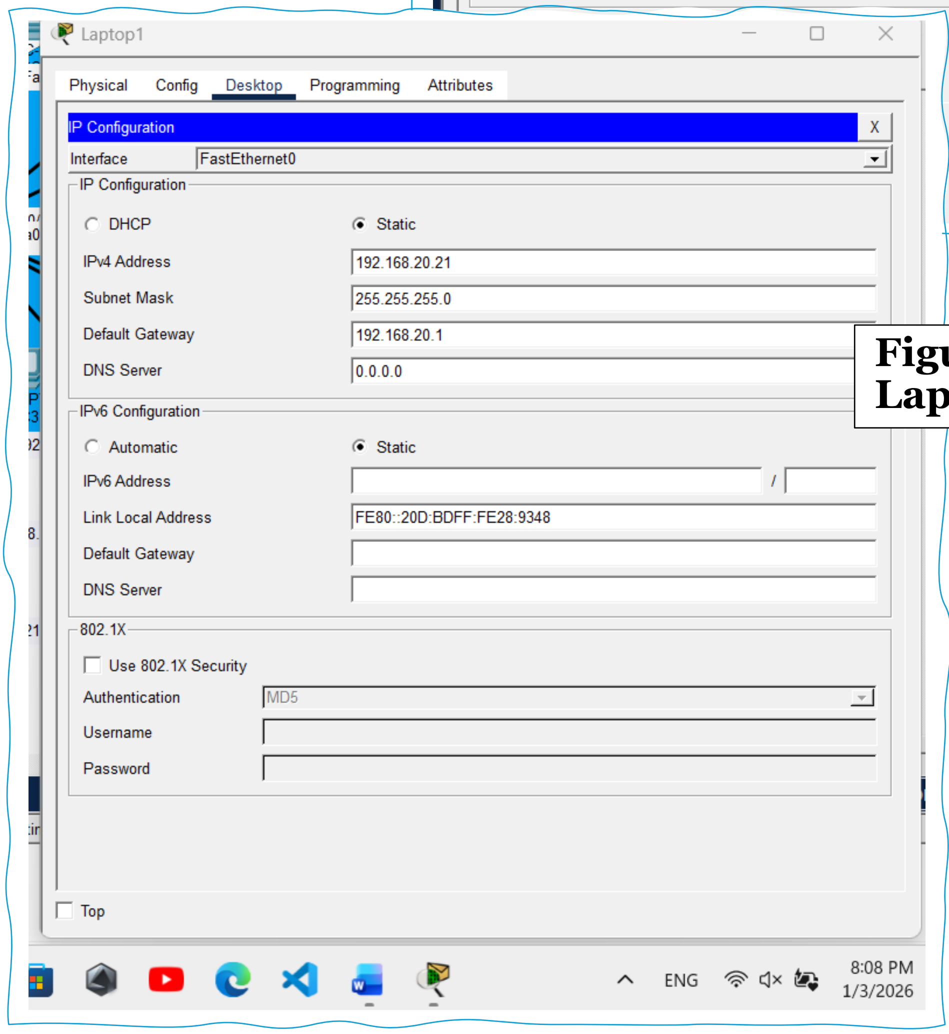
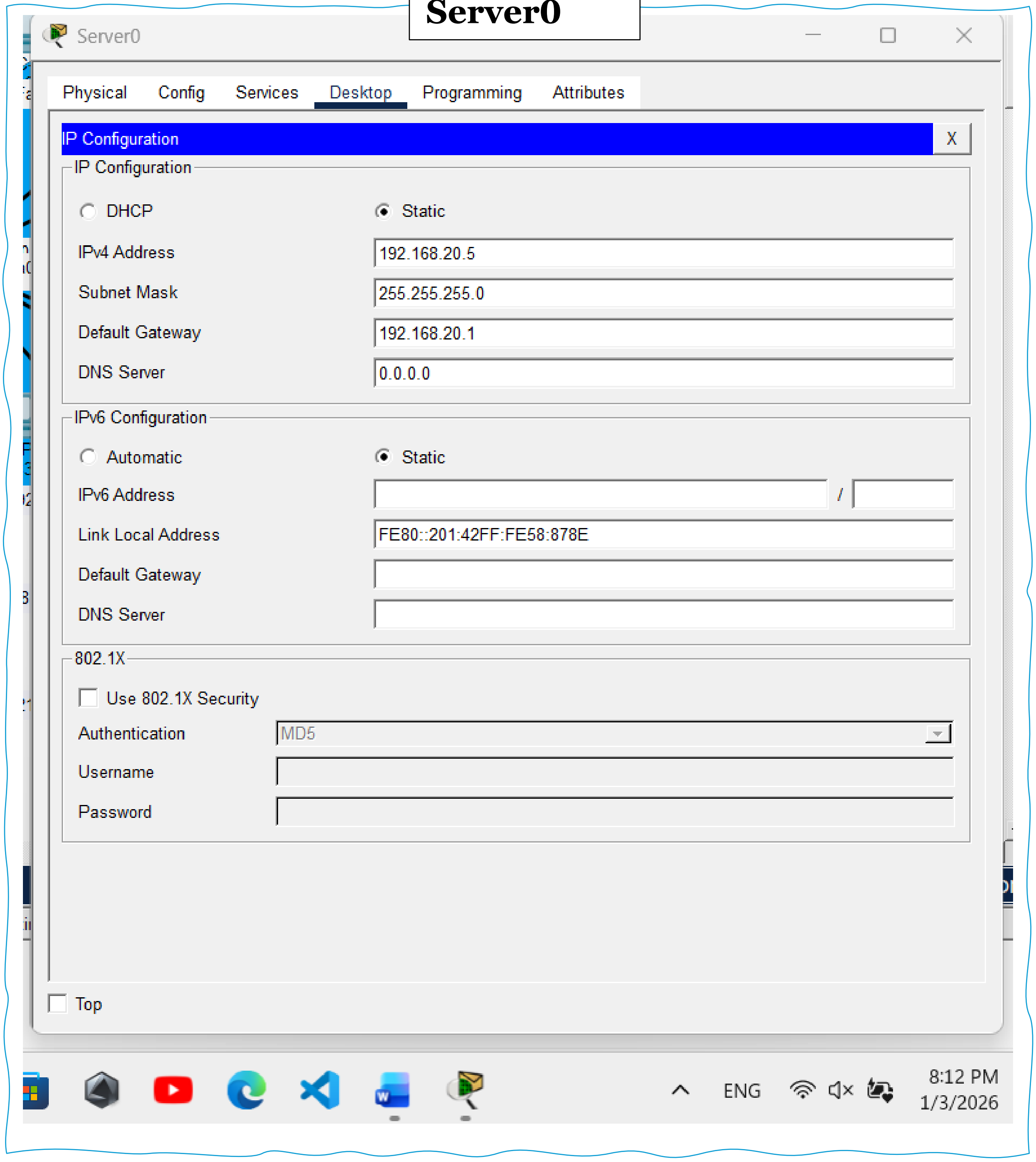


Figure 36: Laptop1

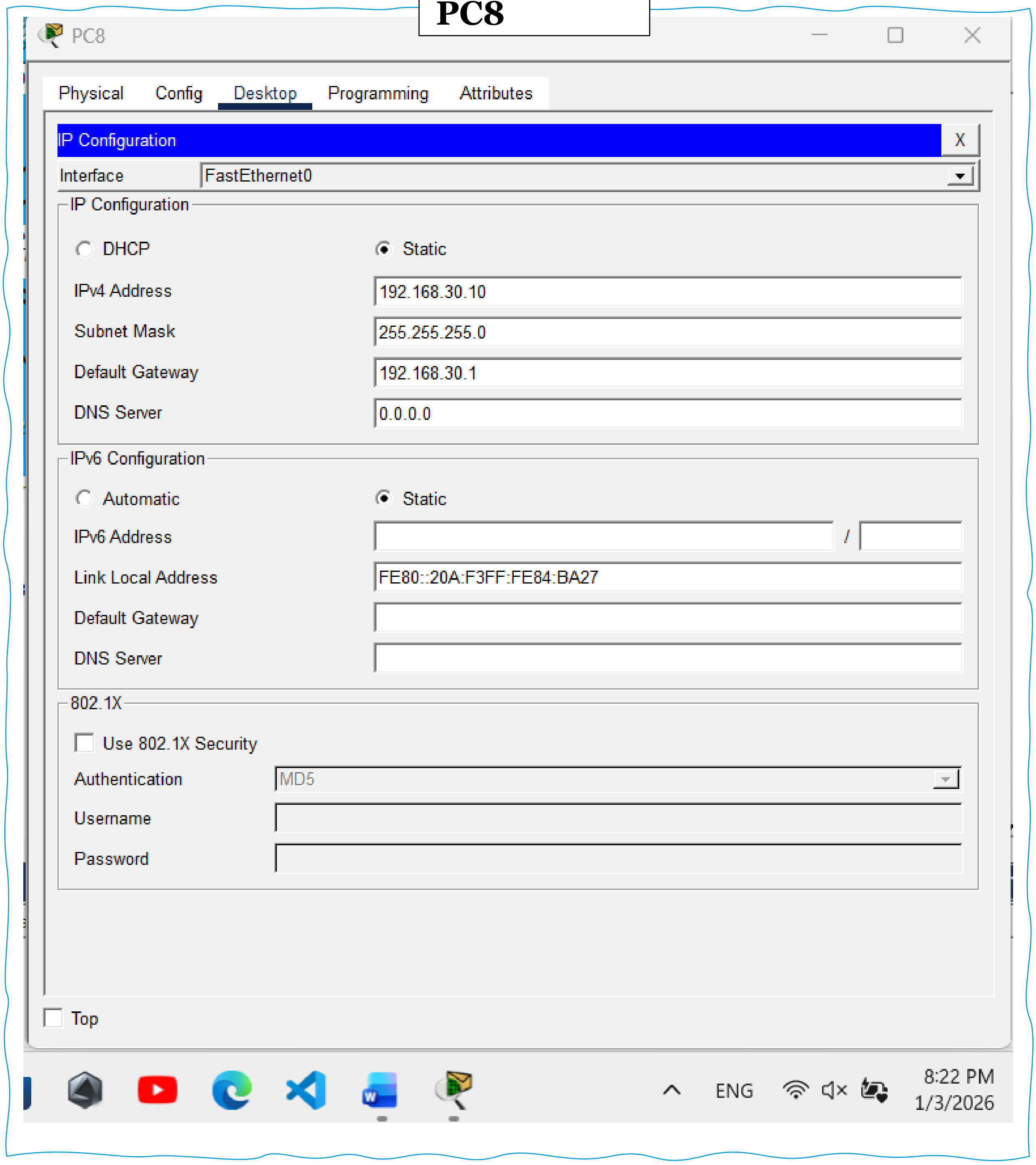
Figure 37: Server0



3 – Second Floor

Devices on the second floor, which includes nursing stations and treatment units, are configured under **VLAN 30**. Each PC and laptop is assigned an IP address from the **192.168.30.0/24** network and uses **192.168.30.1** as the default gateway. This configuration supports smooth communication for nursing staff and patient care systems while maintaining network isolation from other floors.

Figure 38: PC8



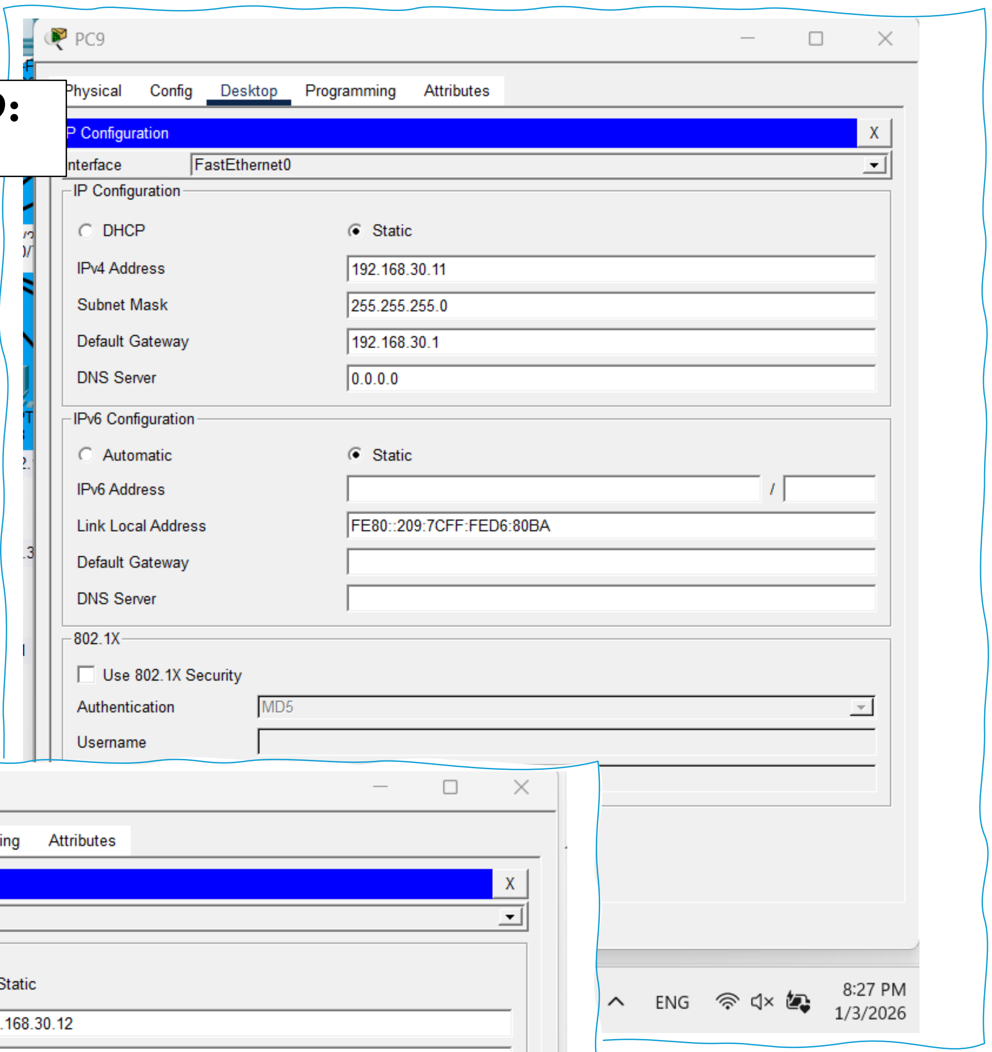


Figure 39: PC9

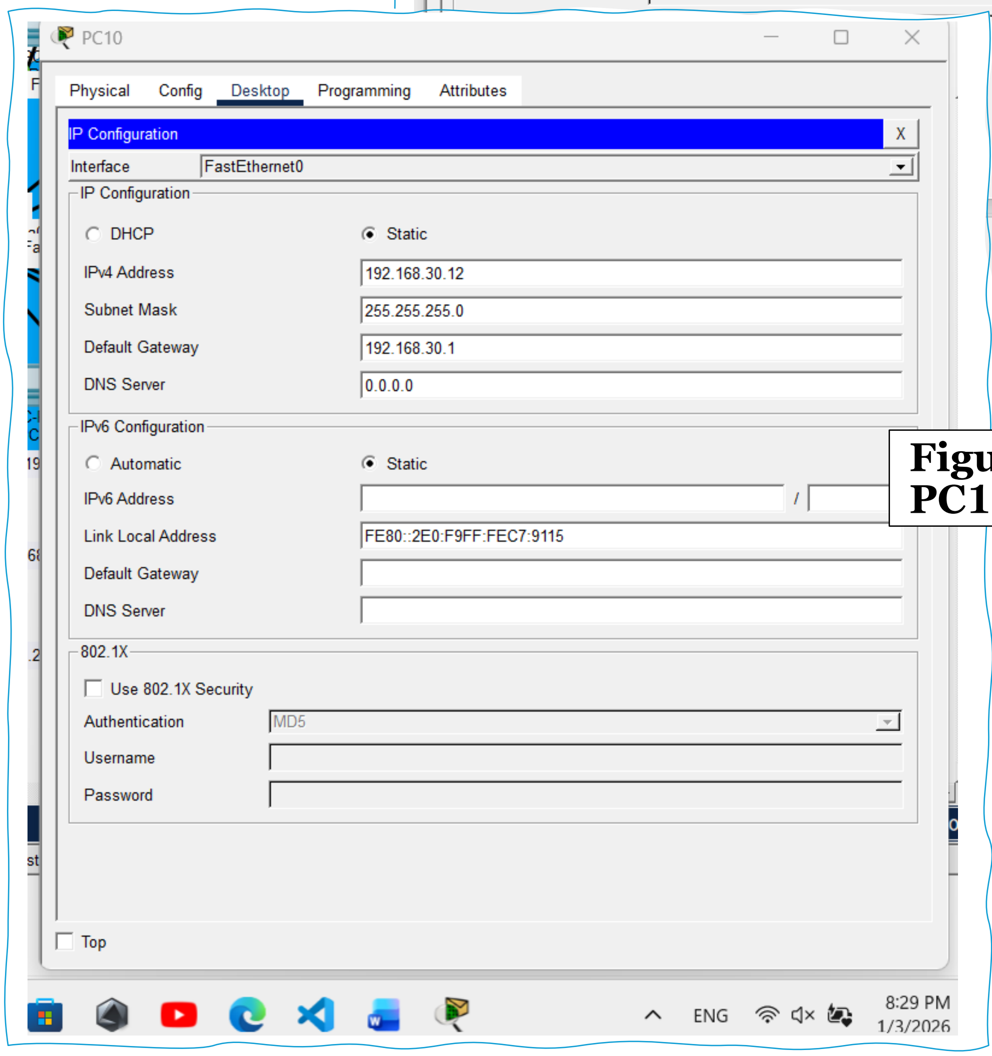
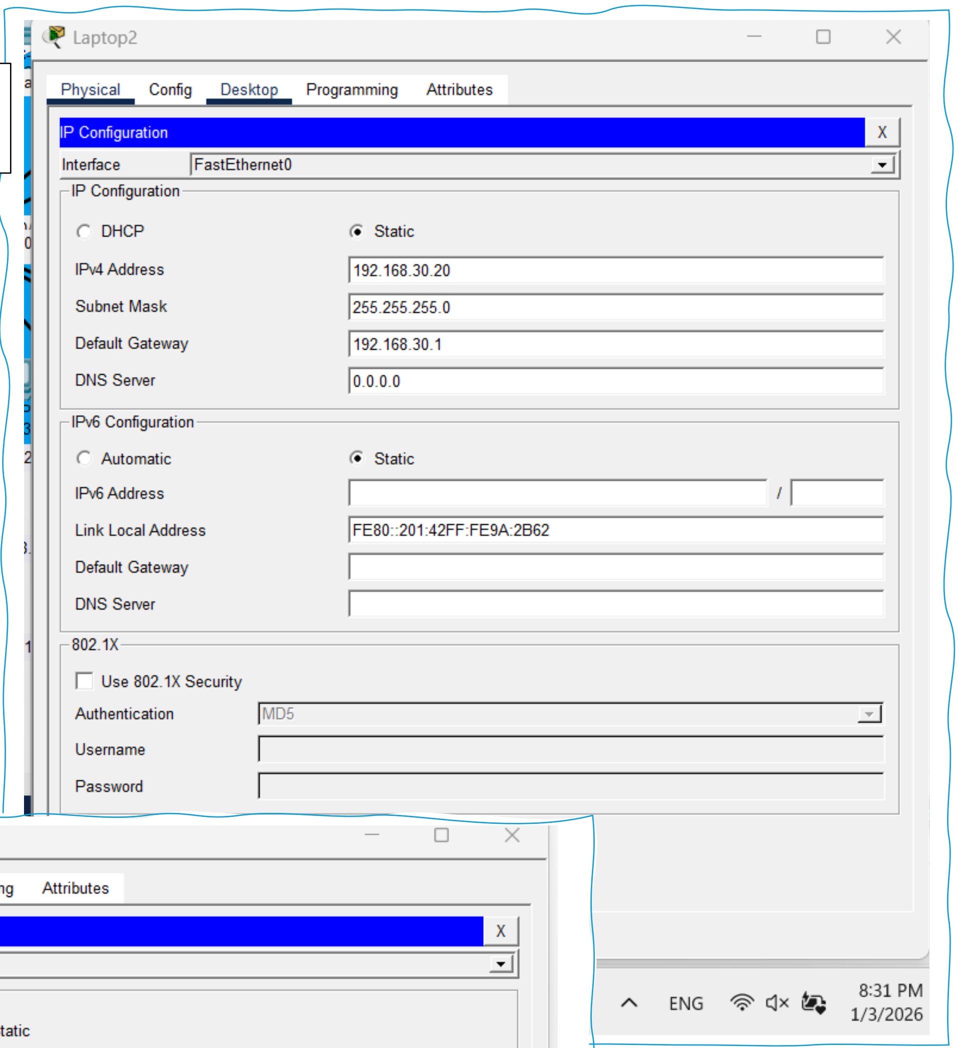


Figure 40: PC10

Figure 41: Laptop2



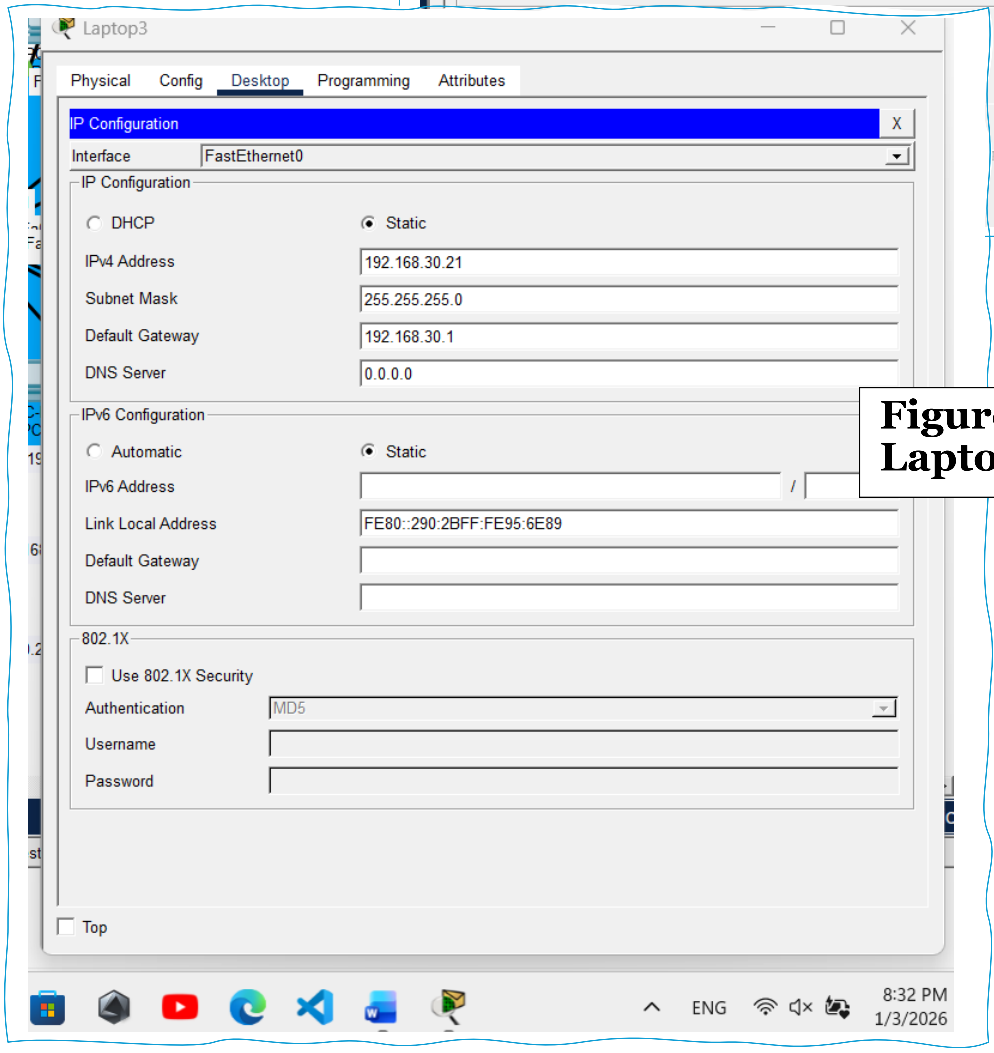
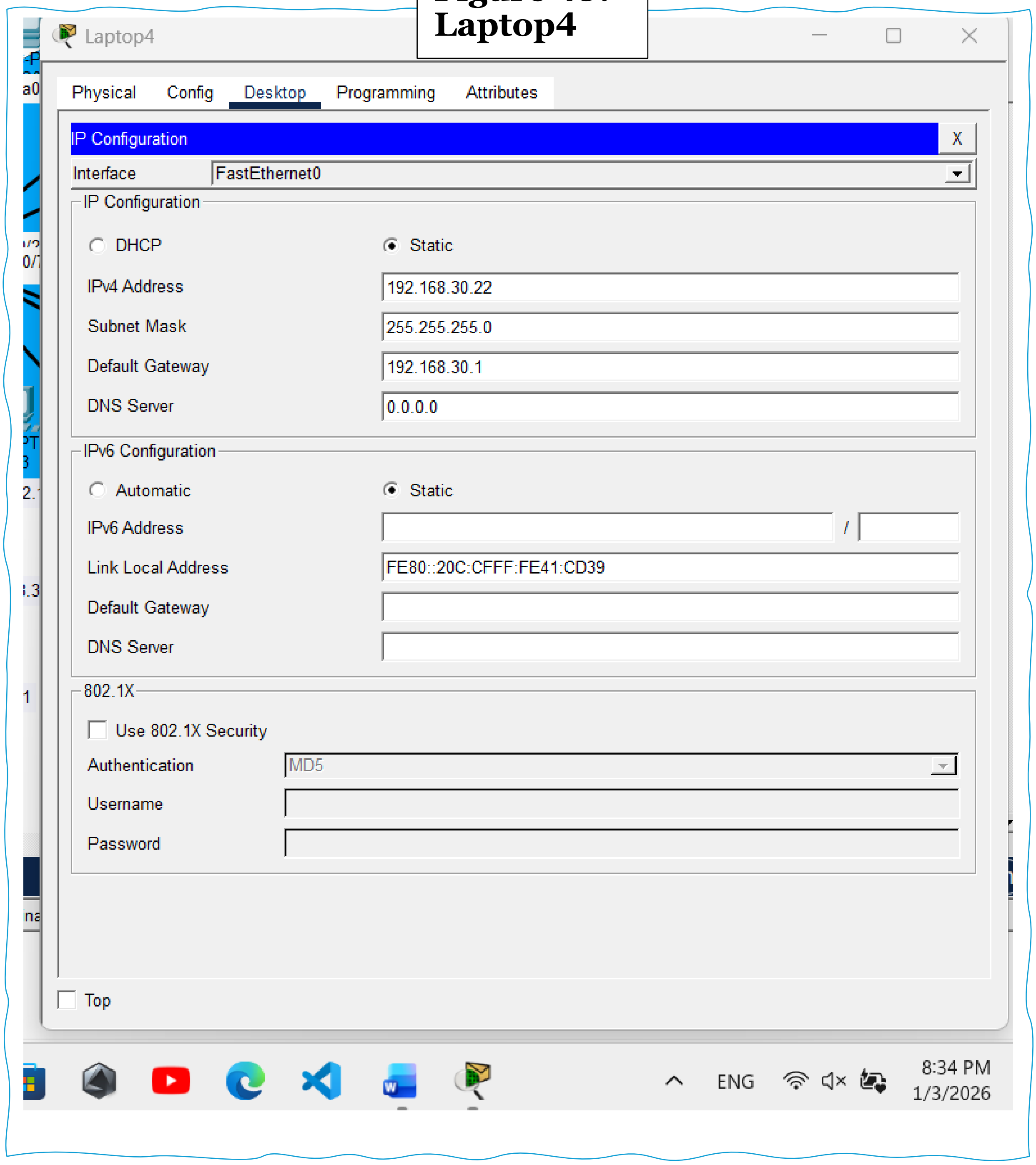


Figure 42: Laptop3

Figure 43: Laptop4



4 - Start Router Inter VLAN configuration step by step

Inter VLAN routing is required because the hospital network is separated into three VLANs (VLAN 10, VLAN 20, and VLAN 30) across different floors. By default, devices inside different VLANs cannot communicate with each other, even if they are connected to the same switches. To solve this, I am configuring the hospital router using the Router on a Stick method, where one physical router interface is divided into multiple sub interfaces. Each sub-interface represents one VLAN and works as the default gateway for that VLAN. This allows secure and controlled communication between floors while keeping the VLAN separation active.

Figure 44:
In switch0 to router a trunk

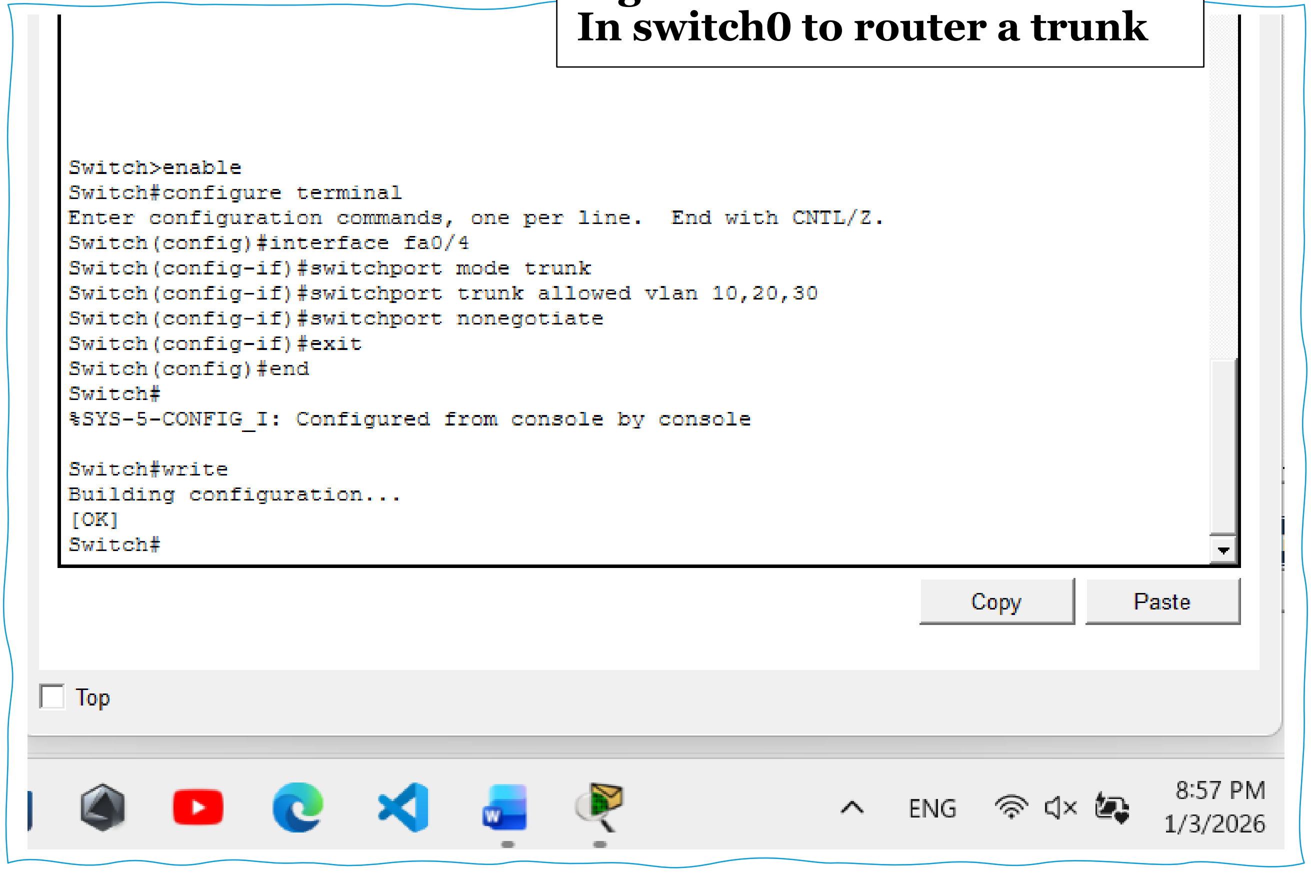
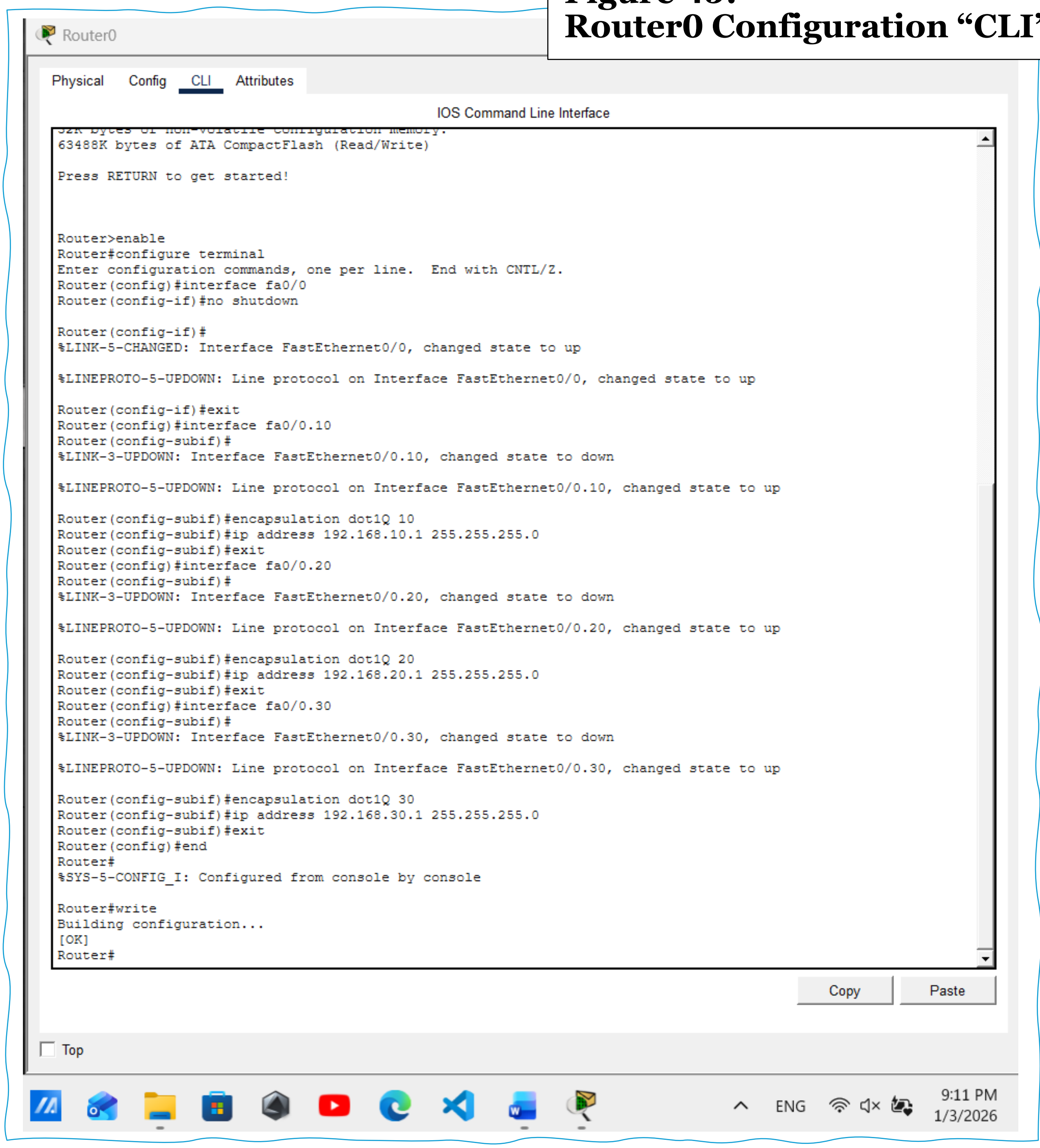


Figure 45:
Router0 Configuration "CLI"



The hospital router is configured to enable communication between different VLANs using the Router-on-a-Stick method. A single physical interface on the router is divided into multiple sub-interfaces, with each sub-interface assigned to a specific VLAN. These sub-interfaces act as the default gateways for VLAN 10, VLAN 20, and VLAN 30, allowing devices on different floors to communicate while maintaining logical network separation. This configuration ensures secure and controlled inter-floor communication and supports efficient data exchange between administrative systems, medical staff devices, and nursing units across the hospital network. *(Donahue, 2011)*

Proving Network Reliability Through Testing and Analysis

Why Testing?

After comPleting the design and implementation of the hospital network, it is important to verify that the system operates as expected. Network testing is carried out to ensure that all devices can communicate correctly within the same VLAN and across different VLANs. Through practical testing methods such as Ping and PDU simulation in Cisco Packet Tracer, the performance, reliability, and correctness of the network configuration are evaluated. This testing phase helps confirm that the implemented VLANs, routing, and services meet the **operational requirements** of the hospital and function efficiently under normal working conditions.

Testing types in PT?

After completing the design and implementation of the hospital network, it is important to verify that the system operates as expected. Network testing is carried out to ensure that all devices can communicate correctly within the same VLAN and across different VLANs. Through practical testing methods such as Ping and PDU simulation in Cisco Packet Tracer, the performance, reliability, and correctness of the network configuration are evaluated. This testing phase helps confirm that the implemented VLANs, routing, and services meet the operational requirements of the hospital and function efficiently under normal working conditions.

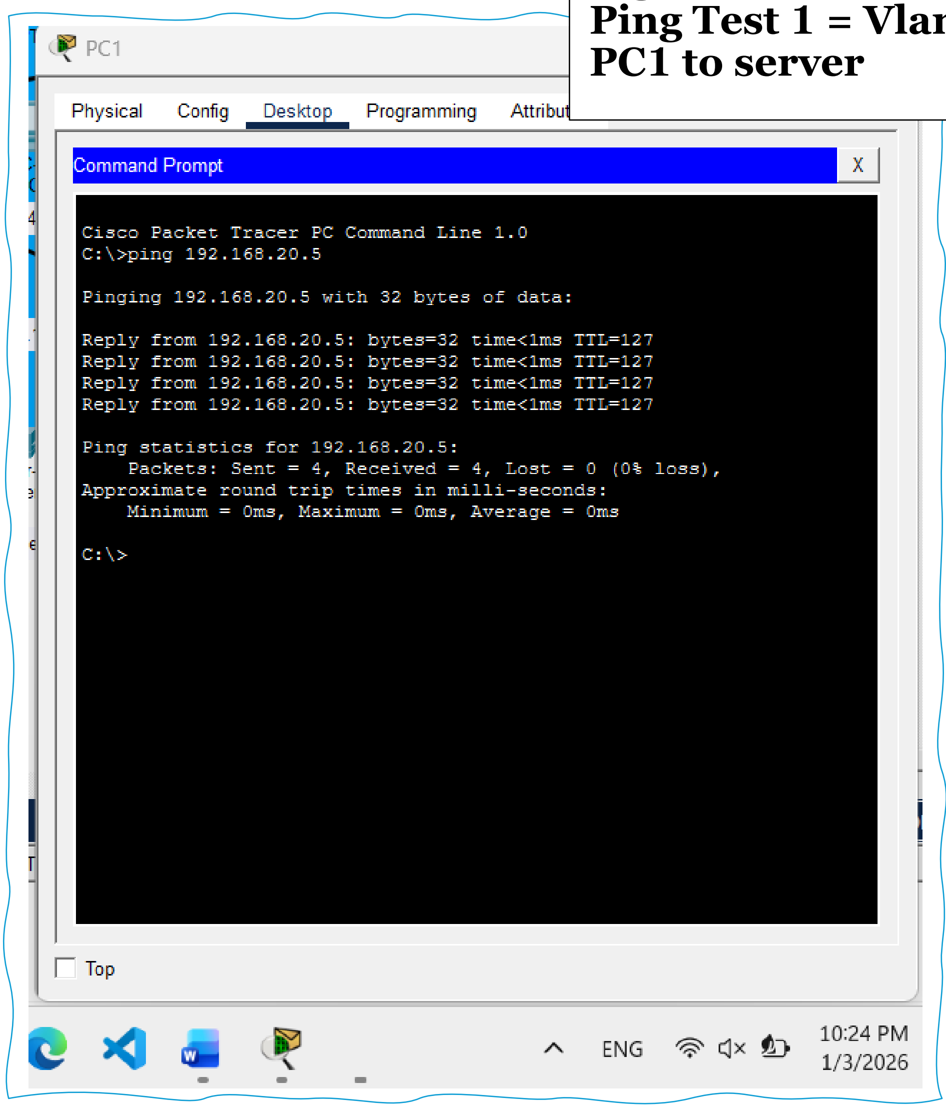
Testing Type	Description	Purpose	How It Is Performed in Packet Tracer	Example in Hospital Network
Ping Testing	Uses ICMP Echo Request and Echo Reply messages to test network connectivity	To confirm that devices can reach each other and that routing is working correctly	Executed through the Command Prompt by sending ping requests to a target IP address	Testing connectivity between a ground floor PC and the first floor server

PDU Simulation	Simulates the transmission of data packets through the network	To visually analyse packet flow and identify routing or configuration issues	Performed in Simulation Mode using the Add Simple PDU tool	Observing packet delivery between devices on different VLANs

1 – Ping testing

Ping testing is used to verify real-time connectivity between network devices by sending ICMP Echo Request messages and receiving Echo Replies. This method helps confirm that devices can reach each other across VLANs using the configured router and switches. In the hospital network, Ping testing is performed between PCs, servers, and devices located on different floors to ensure successful inter-VLAN communication. *(Donahue, 2011)*

Figure 46:
Ping Test 1 = Vlan10 to Vlan 20
PC1 to server



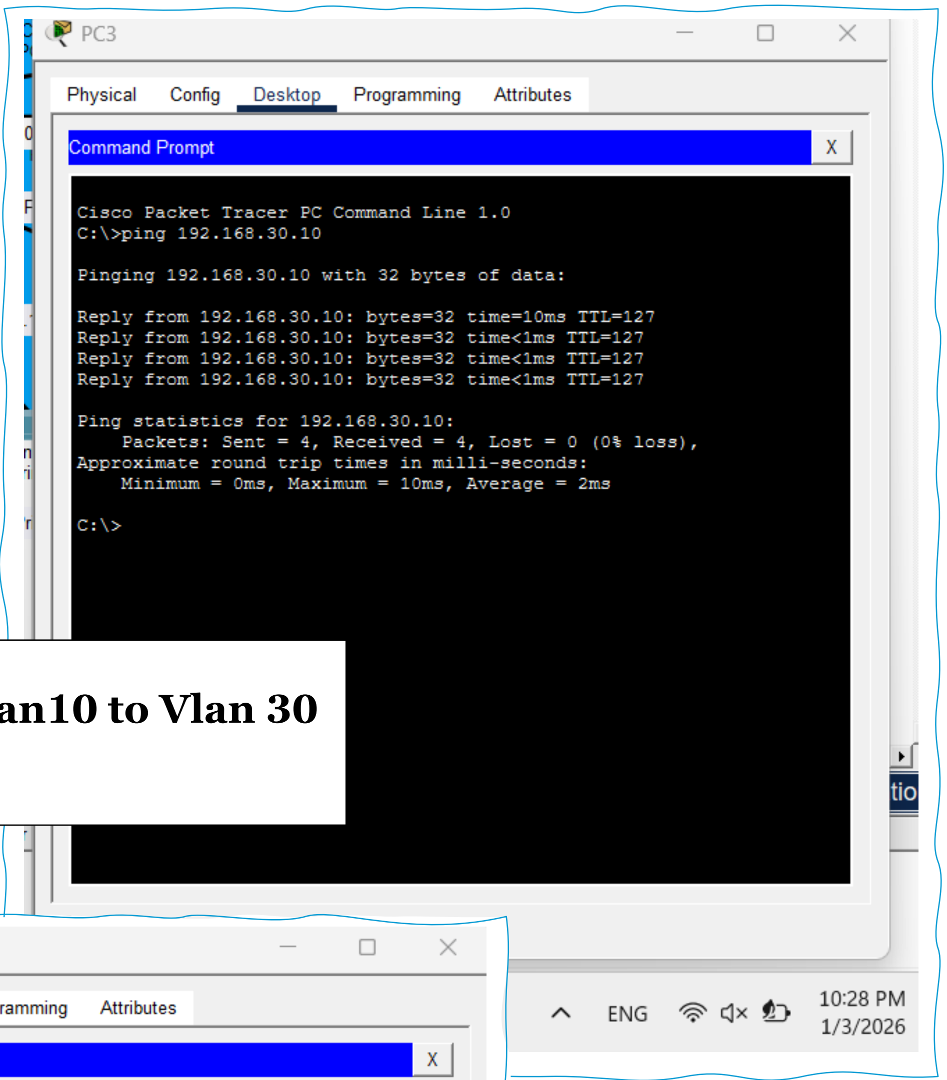


Figure 47:
Ping Test 2 = Vlan10 to Vlan 30
PC3 to PC8

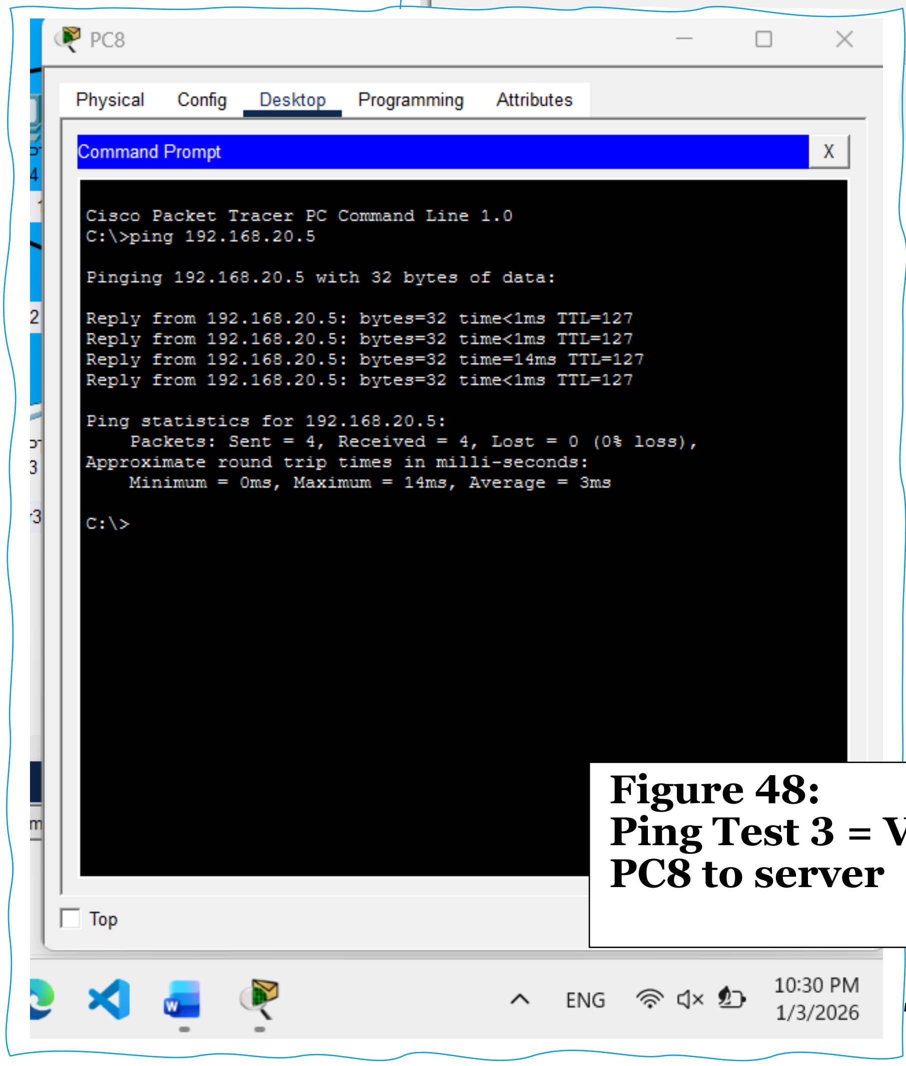


Figure 48:
Ping Test 3 = Vlan30 to Vlan 20
PC8 to server

Ping testing result

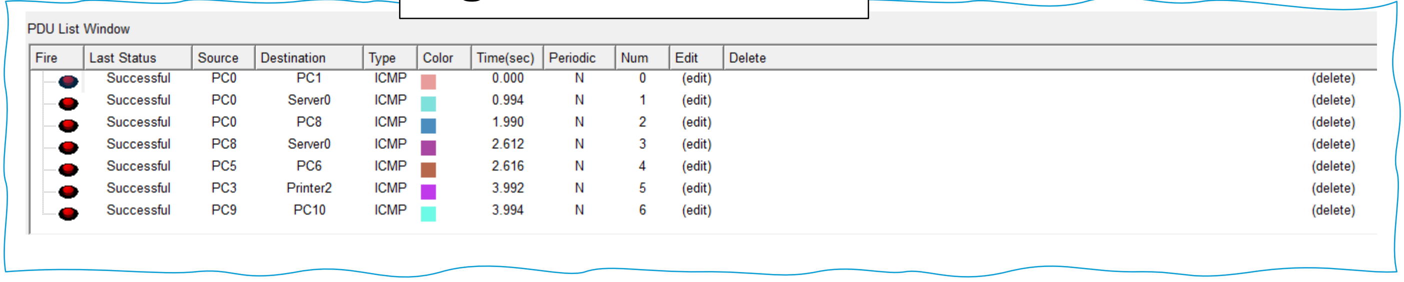
The ping testing carried out on the hospital network confirms **successful connectivity** between devices located on different VLANs and floors. Tests were performed from multiple PCs to verify inter-VLAN communication through the configured router. The results show that **all ping requests received successful replies** with 0% packet loss, indicating that the VLAN configuration and router-on-a-stick implementation are working correctly. The low response times recorded during testing demonstrate efficient routing and stable network performance. These results confirm that the hospital network can reliably support communication between administrative systems, nursing stations, and the central server.

Source Device	Destination IP	Purpose of Test	Result
PC8 (VLAN 30)	**192.168.20.5 (Server)**	Test inter-VLAN communication between nursing floor and server	**Successful**
PC3 (VLAN 10)	**192.168.30.10 (PC8)**	Test inter-VLAN communication between ground floor and second floor	**Successful**
PC1 (VLAN 10)	**192.168.20.5 (Server)**	Test access from ground floor to first floor server	**Successful**

2 – PDU Simulation testing

PDU (Protocol Data Unit) simulation is used to visually analyse how data packets travel through the network. Unlike ping testing, which only confirms connectivity, PDU simulation allows closer observation of packet flow across switches, VLANs, and the router. In the hospital network, PDU simulation is carried out to verify correct VLAN operation, inter-VLAN routing, and proper forwarding of packets between devices on different floors. This testing method helps ensure that the network is correctly configured and that data is transmitted efficiently between PCs, printers, and the central server. ***(Cisco Networking Academy, n.d.)***

Figure 49: PDU Test



PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	Server0	ICMP		0.994	N	1	(edit)	(delete)
	Successful	PC0	PC8	ICMP		1.990	N	2	(edit)	(delete)
	Successful	PC8	Server0	ICMP		2.612	N	3	(edit)	(delete)
	Successful	PC5	PC6	ICMP		2.616	N	4	(edit)	(delete)
	Successful	PC3	Printer2	ICMP		3.992	N	5	(edit)	(delete)
	Successful	PC9	PC10	ICMP		3.994	N	6	(edit)	(delete)

Test	Action	Expected Result	Actual Result	Evidence	Status
01	PDU from PC0 to PC1	Should be successful	Successful	Figure 49	PASS
02	PDU from PC0 to Server0	Should be successful	Successful	Figure 49	PASS
03	PDU from PC0 to PC8	Should be successful	Successful	Figure 49	PASS
04	PDU from PC8 to Server0	Should be successful	Successful	Figure 49	PASS
05	PDU from PC5 to PC6	Should be successful	Successful	Figure 49	PASS
06	PDU from PC3 to Printer2	Should be successful	Successful	Figure 49	PASS
07	PDU from PC9 to PC10	Should be successful	Successful	Figure 49	PASS

PDU simulation was conducted in Cisco Packet Tracer to verify network connectivity within and across different VLANs in the hospital network. The results show that ICMP packets were successfully transmitted between multiple devices, including communication from PC0 to PC1 within the same VLAN, as well as inter-VLAN communication from PC0 to Server0 and PC0 to PC8. Additional tests confirmed successful connectivity between PC8 and Server0, PC5 and PC6 on the first floor, and communication between PCs and printers such as PC3 to Printer2. All PDUs recorded a **successful status** with no packet loss, and response times remained low, indicating efficient routing and switching performance. These results confirm that VLAN configuration, inter-VLAN routing, and device connectivity across all floors are working as expected and meet the operational requirements of the hospital network.

Recommended Enhancements for my Hospital Networked System

Meanwhile the hospital network is now fully designed, implemented, and tested, it is important to think about how this network can be improved in the future. Hospitals are dynamic environments where the number of users, devices, and systems can increase at any time, especially with the use of digital patient records and real-time monitoring systems. Even though the current network meets the operational requirements and performs successfully during testing, there are several enhancements that can further improve security, performance, reliability, and scalability. These recommendations focus on making the network more secure, easier to manage, and ready for future expansion without major redesign.

Mind Organizing table “Enhancement things”

Area	Current Situation	Recommended Enhancement	Benefit to the Network
Network Security	VLANs are implemented to separate departments	Implement Access Control Lists (ACLs) between VLANs	Limits unauthorized access and protects sensitive patient data
Device Management	Devices are accessed locally or without secure remote access	Use SSH instead of Telnet for router and switch management	Provides encrypted communication and improves administrative security
User Authentication	Devices connect without identity-based authentication	Implement user authentication using 802.1X	Ensures only authorized staff can access the network
Network Monitoring	Network status is checked manually	Deploy SNMP-based monitoring tools	Allows real-time monitoring and faster fault detection
Bandwidth Management	All traffic is treated equally	Apply Quality of Service (QoS) for critical systems	Prioritizes medical and patient record traffic for better performance

Fault Tolerance	Single path connections between devices	Introduce redundant links and backup switches	Reduces downtime and improves network reliability
Scalability	Network supports current number of devices	Plan IP addressing and VLAN expansion	Allows easy addition of new floors and departments

(Göransson and Black, 2014)

Why All of this?

The enhancements listed above are designed to **strengthen** the hospital network and prepare it for long-term use. Security-related improvements such as ACLs, SSH, and user authentication help protect sensitive patient information and prevent unauthorized access. This is especially important in a healthcare environment where data confidentiality is critical. Monitoring tools like SNMP provide network administrators with better visibility of network performance, making it easier to identify issues before they affect users.

Performance-related enhancements, such as Quality of Service, ensure that critical hospital systems receive priority over less important traffic, improving response times during peak hours. In addition, introducing redundancy helps maintain network availability even if a device or link fails. Finally, scalability planning ensures that the network can grow smoothly as the hospital expands, without causing disruptions or requiring major redesigns. Overall, these enhancements improve the **efficiency**, **security**, and **reliability** of the networked system. *(Burgess, 2004)*

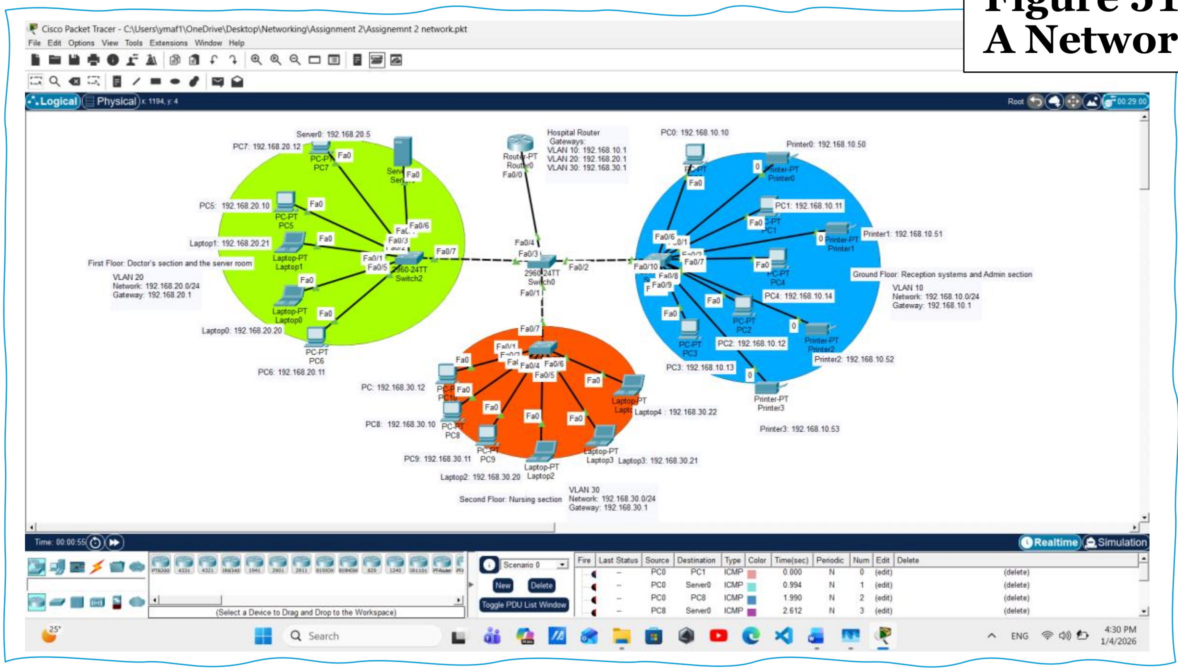
Figure 50

Reflecting on the Design and Performance of my Hospital Network

The hospital network was Designed and implemented with the main aim of providing a secure, reliable, and well-structured communication system across three different floors. Throughout the design and implementation process, careful decisions were made regarding network topology, VLAN segmentation, device placement, and routing methods to ensure that the network could support daily hospital operations effectively. The use of VLANs to separate departments, a core switch to centralize traffic, and a router-on-a-stick configuration for inter-VLAN communication played a key role in achieving this goal.

During implementation, the network was tested using practical tools such as ping testing and PDU simulation in Cisco Packet Tracer. These tests confirmed that devices across different floors could communicate successfully and that services such as the server and FTP were accessible where required. While the overall performance of the network met the expected requirements, reflecting on the design also highlights areas where the system could be improved further. This reflection considers both the strengths and limitations of the implemented network, the decisions made during the design stage, and how proposed enhancements could improve the system's efficiency, security, and scalability in the future.

Figure 51: A Network



Why This Network Design Was Chosen?

The network design was selected based on the **operational** structure of the hospital and the need for clear separation between departments. A **hybrid star topology** was chosen because it allows all floor switches to connect to a central core switch, making the network **easier to manage and troubleshoot**. This design also helps limit the impact of failures, as an issue on one floor does not directly affect the entire network. Using a core switch as the central point **ensures better traffic control** and supports future expansion if additional floors or departments are added. *(Donahue, 2011)*

VLANs were implemented on a per-floor basis to logically **separate network traffic** between administrative staff, medical personnel, and nursing units. This decision improves security by reducing unnecessary access between departments and also enhances performance by limiting broadcast traffic. The router-on-a-stick approach was chosen for inter-VLAN routing as it provides an efficient and cost-effective solution using a single router interface. Overall, this design balances simplicity, performance, and scalability, making it suitable for the hospital environment while remaining flexible for future enhancements.

The good side?

Design Aspect	Design Decision Implemented	Strength Identified	Impact on the Network
Network Topology	Hybrid star topology with a core switch	Centralized management and simplified troubleshooting	Makes it easier to monitor traffic and isolate faults without affecting the entire network
VLAN Implementation	VLANs assigned per floor	Logical separation of departments	Improves security and reduces unnecessary broadcast traffic
Core Switch Usage	Dedicated core switch connecting all floor switches	Efficient traffic handling	Ensures stable communication between floors and supports high network performance
Router-on-a-Stick	Single router interface with multiple sub-interfaces	Cost-effective inter-VLAN routing	Allows secure communication between VLANs without additional hardware

IP Addressing	Structured IP addressing per VLAN	Easy identification and management of devices	Simplifies troubleshooting and future expansion
Server Placement	Central server located on the first floor	Controlled and reliable access to services	Improves availability of hospital services such as FTP and data access
Device Distribution	Devices grouped by function and floor	Organized network layout	Reduces configuration complexity and improves clarity
Testing Approach	Ping and PDU simulation testing	Early detection of configuration issues	Confirms correct implementation and reliable performance
Scalability	VLAN-based structure	Ready for future growth	New devices or departments can be added with minimal changes
Manageability	Clear separation between access and core layers	Easier administration	Reduces operational effort for network maintenance

The limitation side and how enhancements address them

Although the hospital network performs well and meets its current operational requirements, reflecting on the implementation highlights several limitations that could affect the network in the long term. These limitations are mainly related to security depth, scalability, monitoring, and fault tolerance. The current design focuses on functionality and logical separation, but as the hospital grows and more sensitive data and devices are added, additional measures will be required. The proposed enhancements aim to address these limitations by improving security, reliability, and manageability while maintaining the original design structure.

Identified Limitation	Description of Limitation	Proposed Enhancement	How the Enhancement Improves the System
Limited Inter-VLAN Security	VLANs are separated, but there are no access restrictions between them	Implement Access Control Lists (ACLs)	Controls traffic between VLANs and protects sensitive patient data
Basic Device Access	Network devices can be accessed without secure remote management	Use SSH for device management	Encrypts management traffic and prevents unauthorized access
Lack of User Authentication	Devices can connect without identity verification	Implement 802.1X authentication	Ensures only authorized staff can access the network
No Central Monitoring	Network status is checked manually	Deploy SNMP-based monitoring tools	Allows real-time monitoring and faster fault detection
No Traffic Prioritization	All network traffic is treated equally	Apply Quality of Service (QoS)	Prioritizes critical medical and patient record traffic
Single Points of Failure	Core connections rely on single links	Introduce redundancy and backup links	Reduces downtime and improves reliability
Limited Scalability Planning	Network supports current size only	Plan VLAN and IP expansion	Makes future growth easier without redesign

The enhancements proposed above directly address the limitations identified during the reflection process. By improving security through ACLs, SSH, and authentication, the network becomes more suitable for handling sensitive healthcare data. Monitoring and QoS enhancements improve visibility and performance, ensuring that critical systems operate efficiently even during high usage periods. Adding redundancy and scalability planning further strengthens the network, making it more resilient and adaptable to future expansion. Overall, these enhancements build on the existing design rather than replacing it, demonstrating that the original network provides a strong foundation that can be significantly improved through thoughtful upgrades.

Final Judgement on the Implemented Network “and Self Grading”

Reflecting on the overall implementation, the hospital network can be considered **a successful first attempt** at designing and deploying a complete networked system. The network **meets the main requirements** by providing connectivity across all three floors, separating departments using VLANs, and enabling communication between different VLANs through the router. The testing carried out using ping and PDU simulation confirmed that devices, servers, and printers can communicate as expected, showing that the core configuration was implemented correctly.

As this was the first time designing and implementing a network of this scale, some limitations are present, particularly in areas such as advanced security, monitoring, and redundancy. These **limitations do not prevent the network from functioning**, but they highlight areas where further learning and improvement are needed. The recommended enhancements demonstrate how the network could be improved in the future to better suit a real hospital environment and to support growth, higher security requirements, and improved reliability.

Taking into account the successful implementation, testing results, and the learning gained throughout the process, the network would be self assessed at **7.5 out of 10**. This score reflects a strong foundational design for a first time implementation, while also recognising that further development and experience would be required to achieve a more advanced, enterprise level network.

Figure 52

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The End